

HOSPITAL READMISSION CRISIS

**24ADI204 - DATA SCIENCE AND
VISUALISATION**

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*in partial fulfilment for the
award of the degree of*

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



KUMARAGURU COLLEGE OF TECHNOLOGY

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BONAFIDE CERTIFICATE

Certified that this project report **HOSPITAL READMISSION CRISIS** is the Bonafide work of “Anusri T-24BAD010, Guruprasad N - 24BAD029, Shalini S - 24BAD409” who carried out the project work under my supervision.

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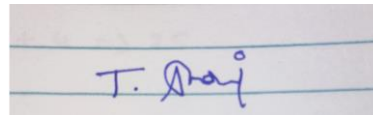
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External Examiner

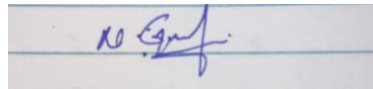
DECLARATION

I affirm that the project work titled **HOSPITAL READMISSION CRISIS** being submitted in partial fulfillment for the award of B. TECH Artificial Intelligence and Data Science is the original work carried out by us. It has not formed the part of any other project work

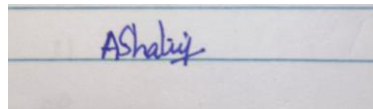
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EXECUTIVE SUMMARY:

This project focuses on analysing hospital readmission patterns among diabetic patients using a real-world healthcare dataset from Kaggle. The main objective was to identify important clinical and demographic factors that influence whether a diabetic patient gets readmitted within 30 days after discharge.

The dataset used for this project contains more than 100,000 patient encounter records and 50 clinical features, including age, race, admission type, diagnosis, medications, laboratory procedures, and readmission status. Since healthcare data often contains missing values, categorical inconsistencies, and outliers, a structured data preprocessing workflow was performed.

The workflow began with data cleaning, including replacing invalid values, removing highly missing columns, handling unknown categories, and generating a binary target variable for readmission within 30 days. Outliers in numerical features were treated using the Interquartile Range (IQR) method. Exploratory Data Analysis was then used to understand patient demographics, hospital stay patterns, and treatment characteristics.

Further, data transformation techniques such as label encoding and feature scaling were applied to prepare the dataset for advanced analysis. Correlation analysis and feature selection helped identify the most influential features affecting readmission. Principal Component Analysis (PCA) was also performed to reduce dimensionality and visualize major patterns.

Key findings indicate that factors such as: number of inpatient visits, emergency visits, time in hospital, number of diagnoses, and treatment complexity significantly affect readmission risk.

Final insights:

Diabetic readmission is mainly influenced by combined factors such as past hospital visits, treatment complexity, and length of stay. Identifying these patterns early can

help hospitals reduce avoidable readmissions through better follow-up care.

This project demonstrates how data visualization and analytical storytelling can

support healthcare decision-making and improve overall patient care strategies

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
EDA	Exploratory Data Analysis
PCA	Principal Component Analysis
IQR	Interquartile Range
Z – SCORE	Standard Score
CSV	Comma Separated Values
KDE	Kernel Density Estimation

LIST OF FIGURES

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HOSPITAL READMISSION CRISIS

1.INTRODUCTION

1.1 Background of the Problem

Hospital readmission is a major healthcare challenge, especially for diabetic patients who require continuous monitoring and long-term treatment. Frequent readmissions increase treatment costs, hospital workload, and patient health risks.

1.2 Importance of the Study

Hospital readmission among diabetic patients is a critical issue in healthcare because it reflects both patient health outcomes and the quality of hospital care. Unplanned readmissions increase treatment costs, place additional burden on healthcare systems and may indicate gaps in discharge planning or follow-up care.

This study is important because it helps understand the major clinical and operational factors that contribute to readmission. By analyzing these patterns, hospitals can make better decisions, improve resource allocation, and provide timely interventions for high-risk patients. The study also highlights the practical value of data visualization in solving real-world healthcare problems.

1.3 Objectives of the Project

- To analyse hospital readmission patterns among diabetic patients.
- To identify factors influencing readmission.
- To clean and preprocess healthcare data effectively.
- To visualize hidden trends and patterns.
- To apply PCA for dimensionality reduction and better interpretation.

1.4 Scope of the Project

Analytical Scope:

This project focuses on identifying patterns and trends in diabetic patient readmission using exploratory data analysis, correlation analysis, and dimensionality reduction techniques.

Technical Scope:

The project covers data preprocessing, missing value handling, outlier treatment, feature encoding, scaling, and PCA using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

Application Scope:

The insights from this project can support hospitals in identifying high-risk patients, improving discharge planning, enhancing follow-up care, and reducing avoidable readmissions.

2.DATASET DESCRIPTION

2.1 Source of Dataset

Dataset Source: Kaggle – Diabetes 130-US Hospitals Dataset

Dataset Link: <https://www.kaggle.com/datasets/brandao/diabetes>

2.2 Dataset Size

- Total Rows: 101,766 patient encounter records
- Total Columns: 50 clinical and administrative features
- Data Type: Mixed (Numerical and Categorical)
- Time Period Covered: 1999 to 2008
- Data collected from: 130 US hospitals and integrated delivery networks

This large-scale dataset provides a realistic representation of diabetic patient treatment history, hospital visits, and discharge outcomes over a ten-year period.

2.3 Description of Features

Important features include:

- race: Patient ethnicity
- gender: Patient gender
- age: Age group
- time_in_hospital: Length of stay
- num_lab_procedures: Number of lab tests performed
- num_procedures: Number of medical procedures performed
- num_medications: Number of medications prescribed
- number_outpatient: Number of outpatient visits
- number_diagnoses: Total diagnoses recorded
- A1Cresult: Indicates blood sugar control level
- max_glu_serum: Maximum glucose serum test result
- insulin: Insulin medication status
- change: Whether medication was changed during stay

- diabetesMed: Whether patient was on diabetes medication
- diag_1, diag_2, diag_3: Primary, secondary, and tertiary diagnosis codes
- readmitted: Readmission status

2.4 Initial Data Issues

- Missing values represented as '?'
- High missing values in weight, payer_code, and medical_specialty
- Presence of duplicate patient encounters across different visits
- Categorical inconsistencies and unknown labels
- Presence of outliers in numerical columns such as hospital stay and medication count
- Class imbalance in readmission categories
- High dimensionality due to multiple diagnosis and treatment-related features

Despite these challenges, the dataset is highly valuable because it captures detailed patient-level healthcare interactions, making it suitable for identifying readmission patterns and generating practical healthcare insights.

3.DATA PREPROCESSING AND CLEANING

3.1 Data Loading and Inspection

The dataset was imported into the Python environment using the Pandas library. Initial inspection was performed to understand the structure, data types, and quality of the dataset.

The following methods were used:

- `head()` to view sample records
- `shape` to check dataset dimensions
- `info()` to inspect data types and null values
- `describe()` to summarize numerical features
- `value_counts()` for important categorical variables
- missing value summary to identify incomplete fields

This initial inspection helped identify data quality issues and guided the cleaning strategy.

3.2 Handling Missing Values

Steps performed:

- Replaced '?' values with NaN.
- Dropped `weight` and `payer_code` columns due to excessive missing values.
- Filled missing values in `race` and `medical_specialty` with 'Unknown'.

HANDLE THE MISSING AND CATAGERICAL FEATURES

-----WEEK - 3-----

```
01... df.replace('?', np.nan, inplace=True)
```

01...

	encounter_id	patient_nbr	race	gender	age	weight	admission_type_id	discharge_disposition_id	admission_source_id	time_in_hospital	
	0	2278392	8222157	Caucasian	Female	[0-10)	NaN	6	25	1	1
	1	149190	55629189	Caucasian	Female	[10-20)	NaN	1	1	7	3
	2	64410	86047875	AfricanAmerican	Female	[20-30)	NaN	1	1	7	2
	3	500364	82442376	Caucasian	Male	[30-40)	NaN	1	1	7	2
	4	16680	42519267	Caucasian	Male	[40-50)	NaN	1	1	7	1
	...	...	...	...	...	...	...	...	...	...	...
	101761	443847548	100162476	AfricanAmerican	Male	[70-80)	NaN	1	3	7	3
	101762	443847782	74694222	AfricanAmerican	Female	[80-90)	NaN	1	4	5	5
	101763	443854148	41088789	Caucasian	Male	[70-80)	NaN	1	1	7	1

```
df.drop(columns=["weight", "payer_code"], inplace=True, errors='ignore')
```

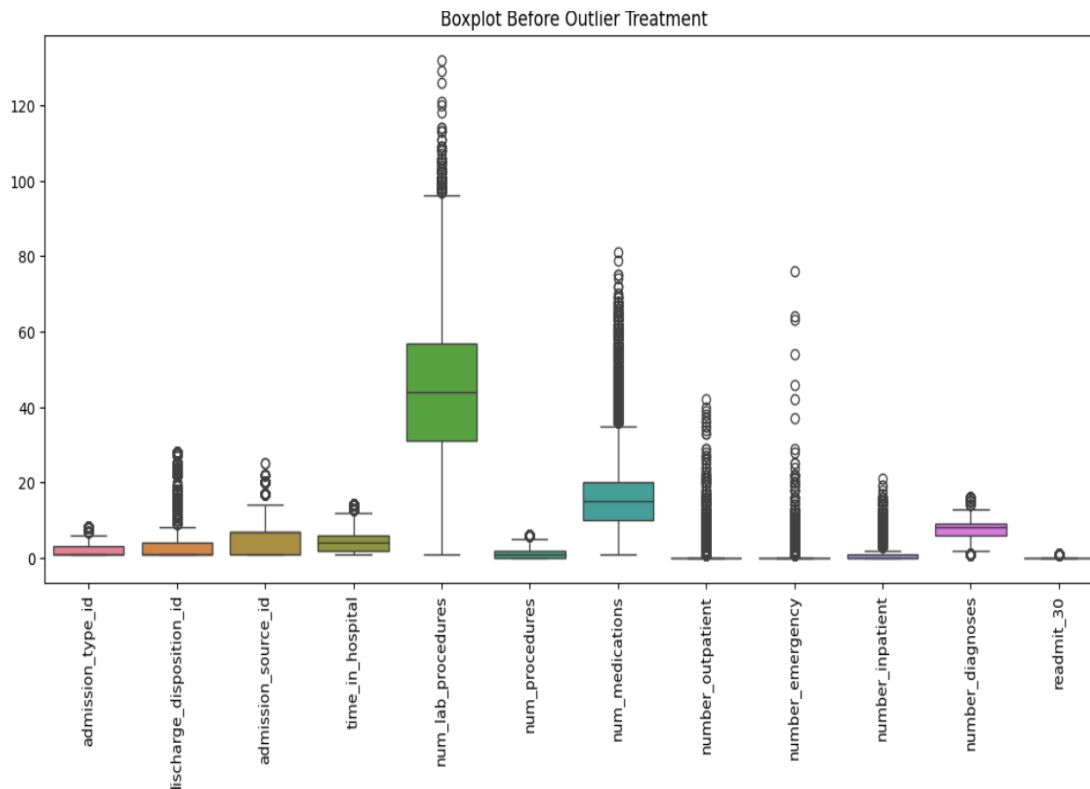
```
df["race"] = df["race"].fillna("Unknown")
df["medical_specialty"] = df["medical_specialty"].fillna("Unknown")
```

Justification:

This improved data completeness without introducing bias through arbitrary imputation.

3.3 Outlier detection and Analysis

Outliers were identified using statistical techniques such as Z-score and Interquartile Range (IQR).



- Wide variation:
 - Features like num_lab_procedures and num_medications have large variability.
- Categorical encoded variables:
 - Features such as admission_type_id, discharge_disposition_id, and admission_source_id have smaller ranges but still contain some unusual values.

3.4 Data Quality Report Summary

The Data Quality Report summarizes the condition of the dataset after preprocessing, ensuring it is clean, consistent, and suitable for further analysis.

Completeness

- Missing values were identified using Pandas
- Imputation techniques (mean, median, mode) were applied
- Columns with excessive missing values were removed

Result: Missing data significantly reduced.

Consistency

- Standardized column names and formats
- Unified categorical values (e.g., similar labels merged)

Result: Improved uniformity across dataset.

Accuracy

- Checked for unrealistic or incorrect values
- Validated numerical ranges and categories

Result: Increased data reliability.

Outlier Treatment

- Detected using Z-score and IQR methods
- Treated using removal or capping

Result: Reduced data distortion.

4.Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is an essential step in the data science workflow that helps in understanding the structure, distribution, and relationships within the dataset. In this project, EDA was performed to identify trends, detect anomalies, understand patient characteristics, and discover patterns related to hospital readmission.

EDA helps transform raw healthcare data into meaningful insights by using statistical summaries and visualizations. It supports better decision-making in later stages such as feature engineering, dimensionality reduction, and predictive analysis.

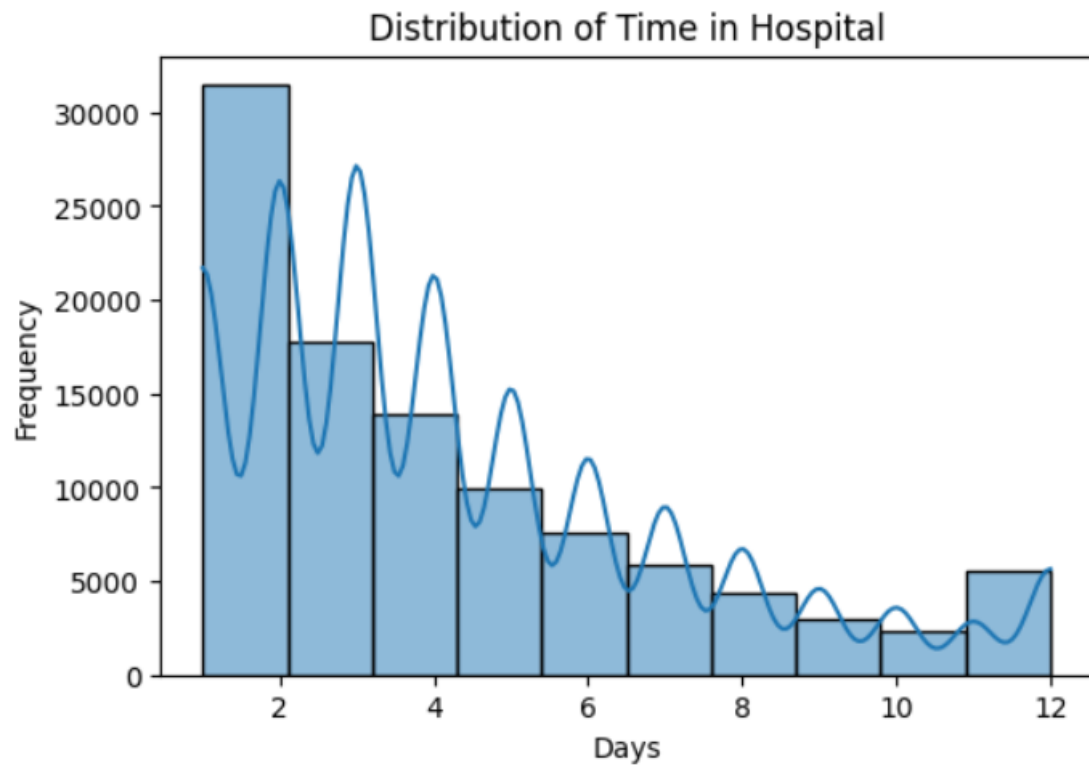
4.1 Univariate Analysis (DISTRIBUTION ANALYSIS)

Univariate analysis focuses on studying one variable at a time to understand its distribution, central tendency, spread, and frequency patterns. It helps identify skewness, outliers, and dominant categories in the dataset.

In this project, univariate analysis was used to examine patient demographics, hospital stay duration, medication count, and other important features through histograms, count plots, and boxplots.

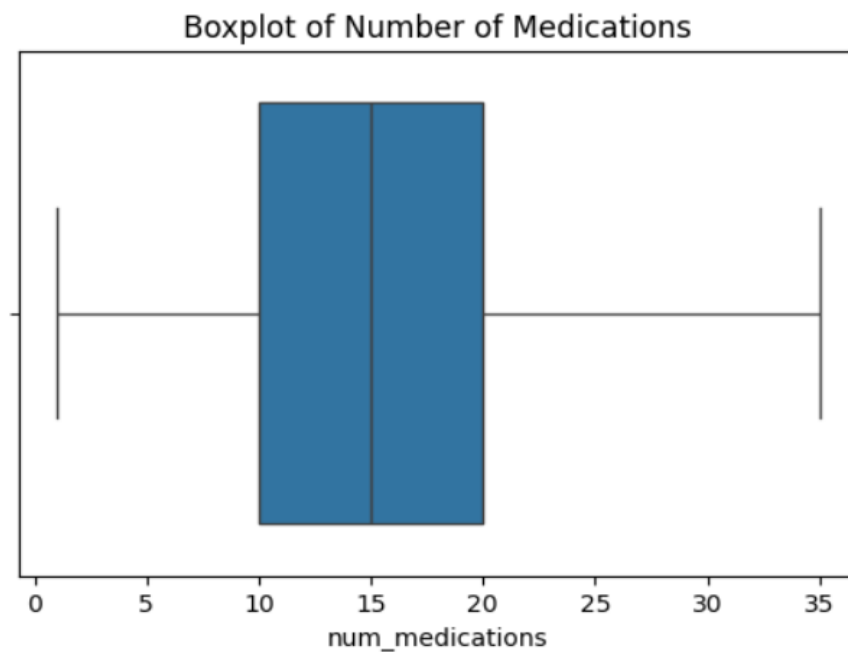
Performed distribution analysis for:

- Time in hospital
- Number of medications
- Readmission
- Gender



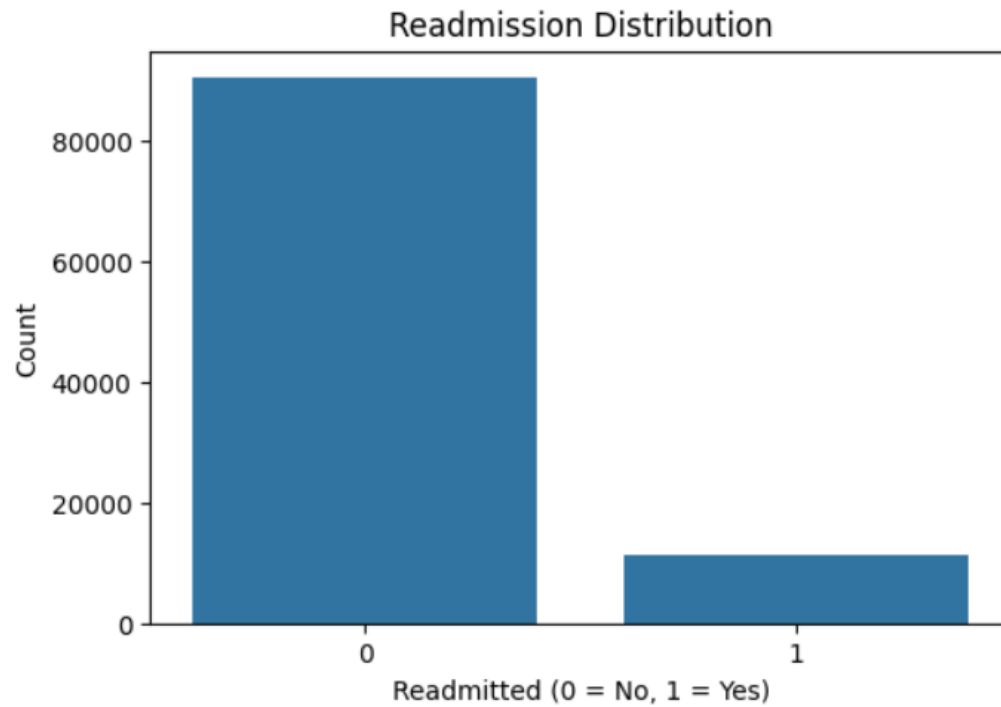
INTERPRETATION:

Most patients stay for a short duration, while only a few have longer hospital stays, indicating that long stays are less common.



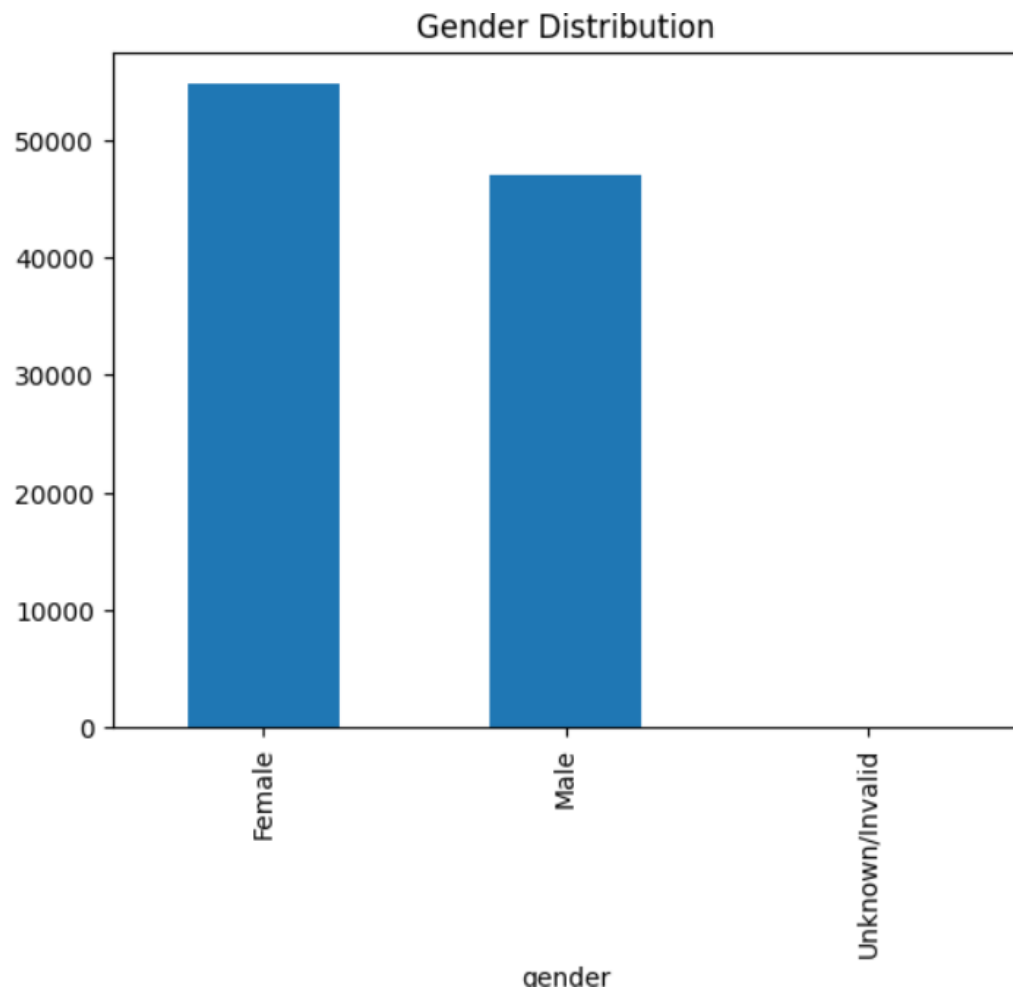
INTERPRETATION:

Some patients take significantly more medications than others, indicating possible severe medical conditions.



INTERPRETATION:

Majority of patients are not readmitted, indicating class imbalance in the dataset.



INTERPRETATION:

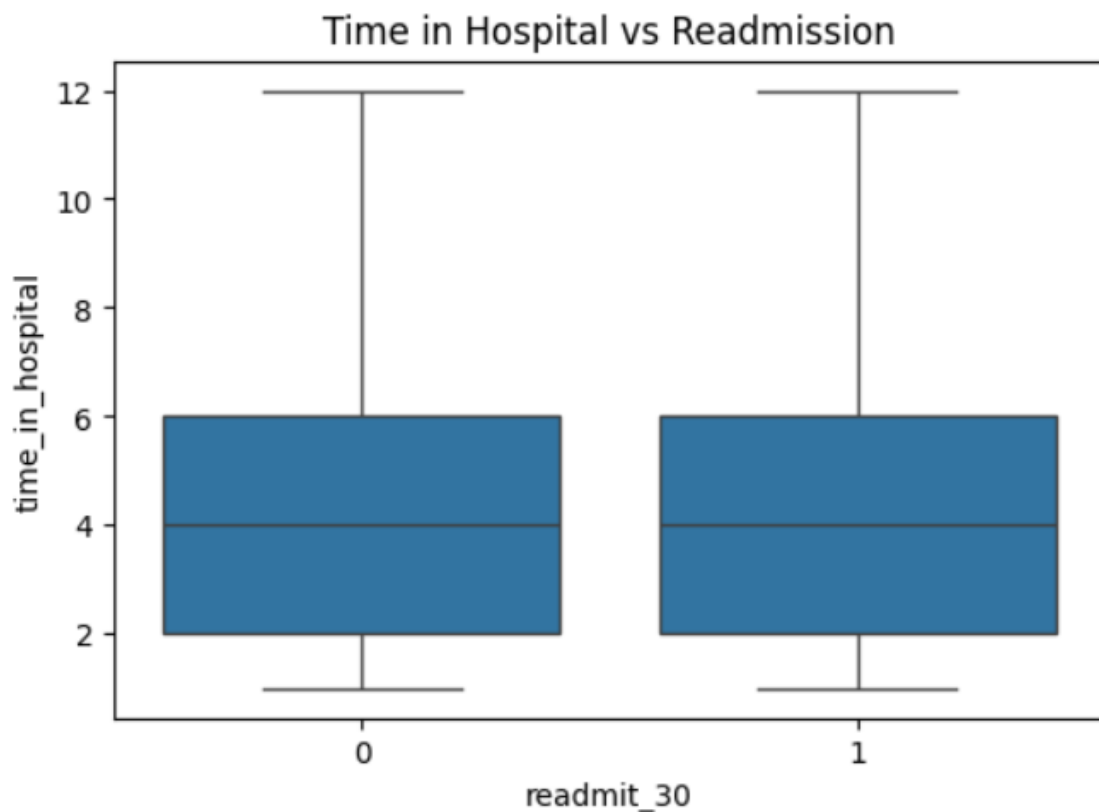
The graph shows the distribution of patients based on gender. It is observed that the number of female patients is slightly higher than male patients.

The “Unknown/Invalid” category has very few entries, indicating minimal missing or incorrect gender data in the dataset.

4.2 Bivariate Analysis (RELATIONSHIP & TREND ANALYSIS)

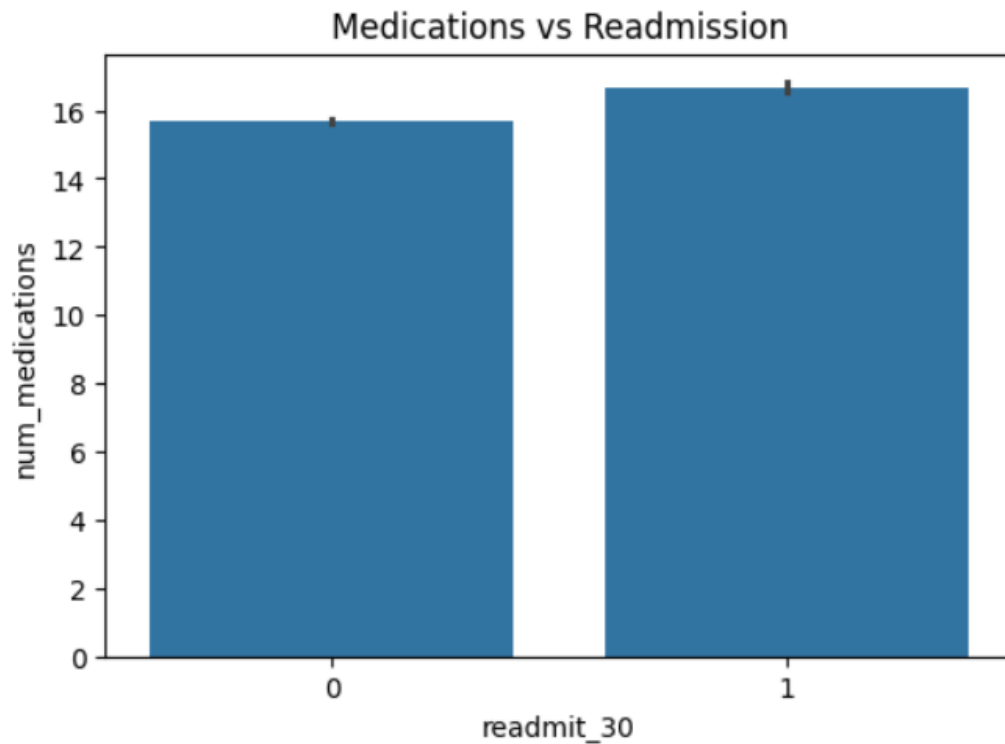
Bivariate analysis examines the relationship between two variables to identify patterns, trends, and possible associations. It is useful for understanding how one feature influences another, especially the target variable.

In this project, bivariate analysis was performed to study the relationship between patient characteristics and readmission status using bar plots, heatmaps, grouped comparisons, and trend analysis.



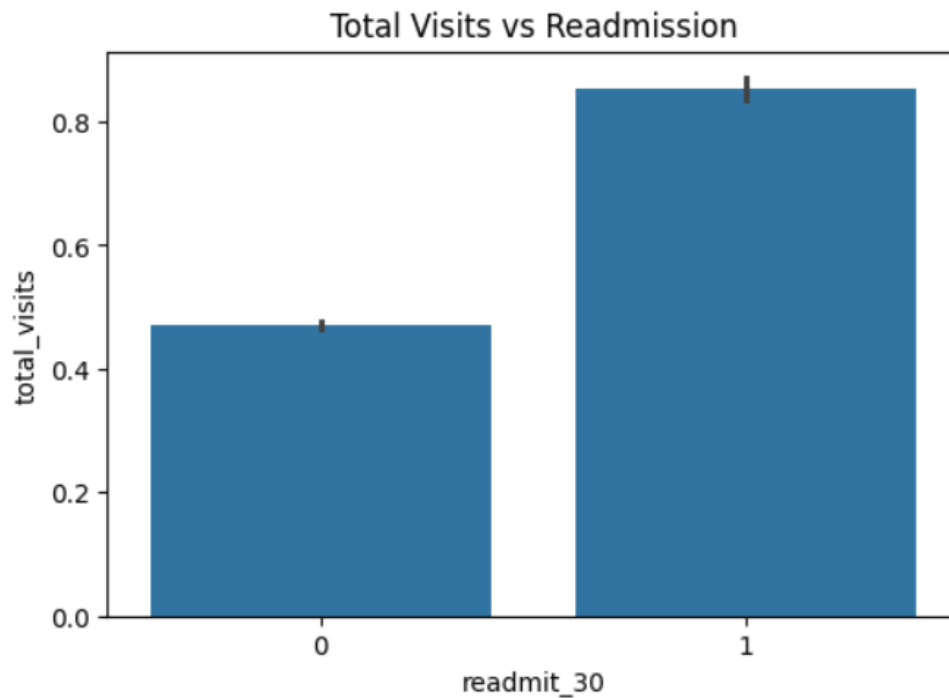
INTERPRETATION:

The boxplot shows that the distribution of hospital stay is very similar for both readmitted and non-readmitted patients. There is no significant difference in median or spread, indicating that hospital stay alone may not strongly influence readmission.



INTERPRETATION:

- The bar chart shows that patients who are readmitted have a slightly higher average number of medications compared to those who are not readmitted. This suggests that patients with more medications may have more complex health conditions, increasing the likelihood of readmission.



INTERPRETATION:

- The bar chart shows that patients who are readmitted have a significantly higher average number of total visits compared to those who are not readmitted. This indicates that frequent hospital visits are strongly associated with a higher likelihood of readmission.

4.4 SUMMARY OF EDA FINDINGS

The exploratory data analysis reveals that hospital readmissions are influenced by identifiable patterns rather than random occurrences. These patterns are observed across multiple factors such as patient demographics, medical conditions, length of hospital stay, and frequency of previous admissions. Patients with longer hospital stays and a higher number of prior visits show a greater likelihood of readmission. Additionally, certain medical conditions and age groups are more prone to repeated hospital visits, indicating vulnerable patient segments.

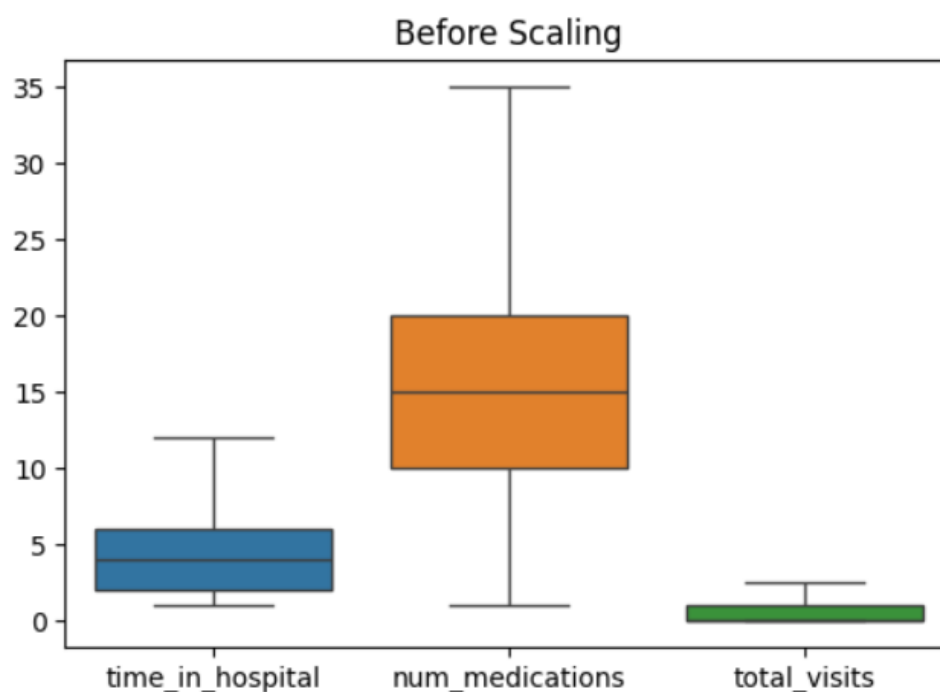
The analysis also highlights that no single feature is sufficient to accurately predict readmission. Instead, a combination of variables—such as diagnosis type, number of procedures, medications, and patient history—must be considered together to capture the complete picture.

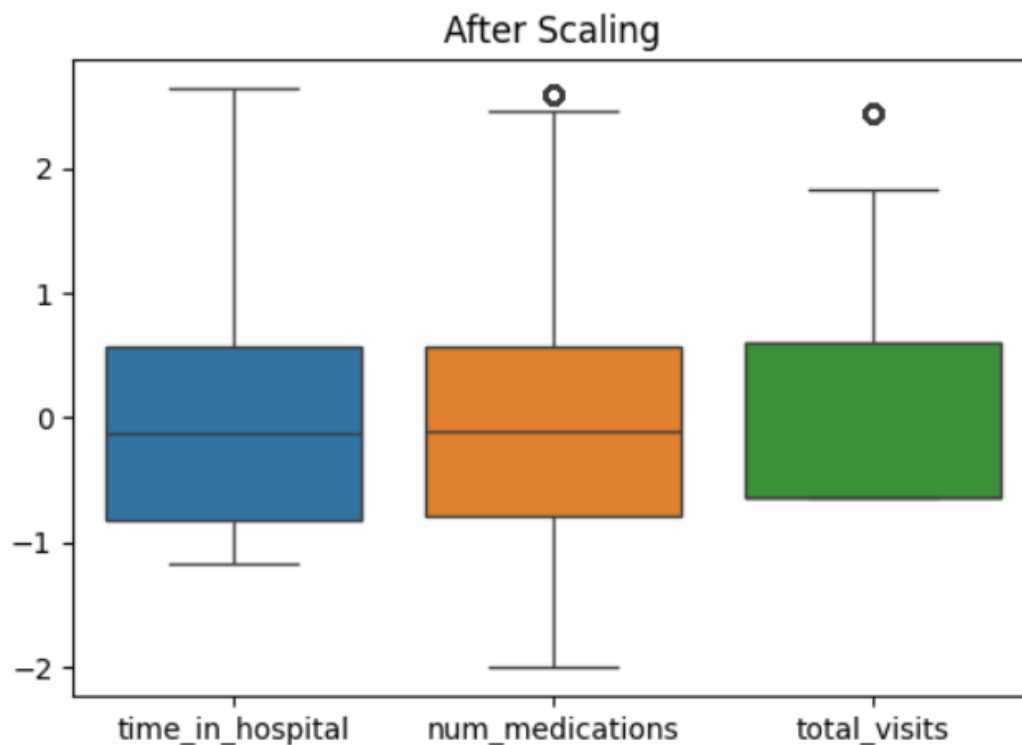
Furthermore, EDA helped identify key trends such as imbalanced class distribution, presence of outliers, and correlations between clinical variables. These insights the preprocessing and feature engineering stages. Overall, the exploratory analysis provides a strong foundation for building effective predictive models and supports better healthcare decision-making aimed at reducing hospital readmission rates.

5.DATA TRANSFORMATION AND FEATURE ENGINEERING

5.1 FEATURE SCALING

Feature scaling is a crucial preprocessing step in the hospital readmission analysis, as dataset contains multiple numerical variables with different units and ranges, such as number of lab procedures, medications, time in hospital, and number of outpatient. Without scaling, features with larger magnitudes may dominate the model, leading to biased predictions.





INTERPRETATION:

Before scaling, features have different ranges and spread. After scaling, all features are normalized to a similar scale, reducing the effect of outliers and improving comparability.

5.2 ENCODING TECHNIQUES

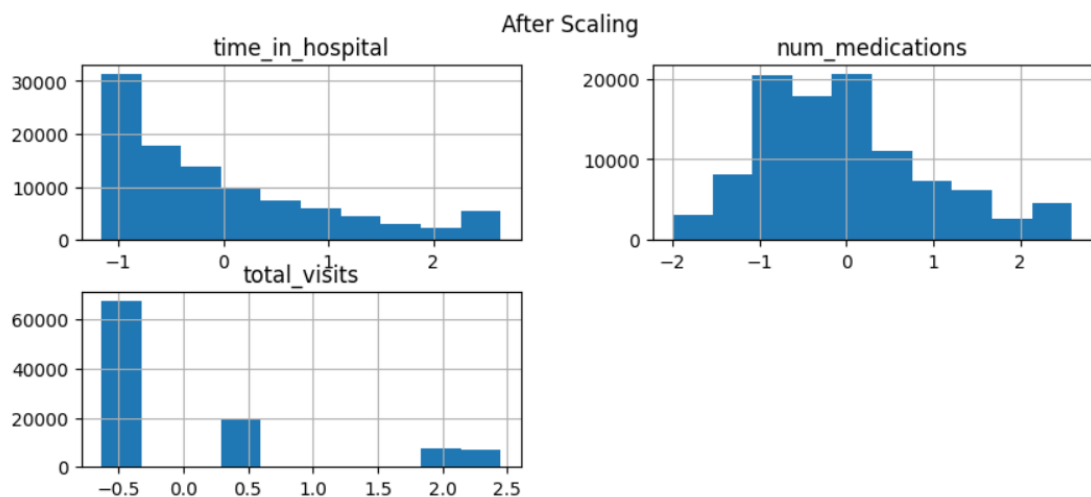
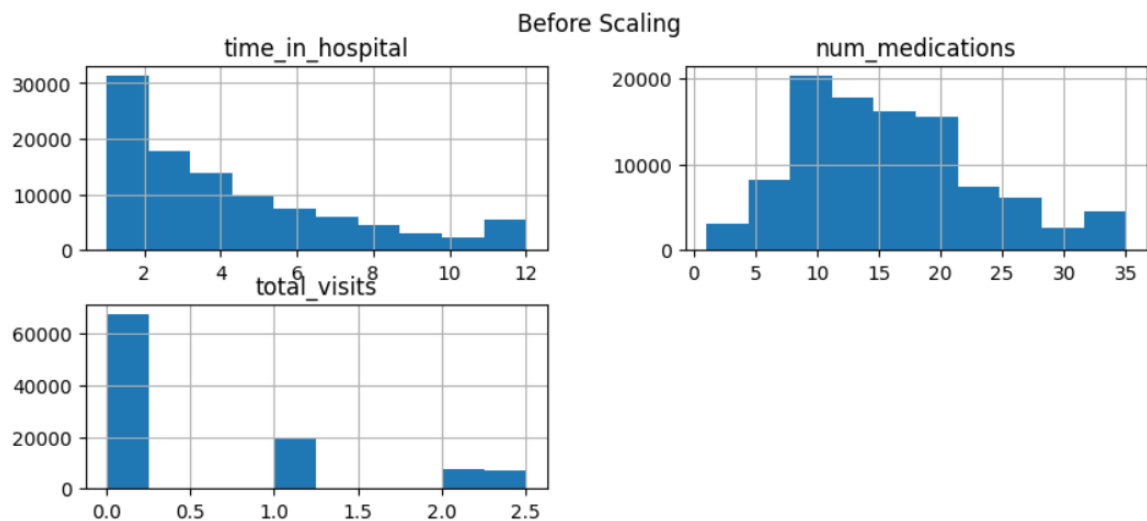
Categorical data converted into numerical form:

- Label Encoding:
 - Assigns integer values
 - Suitable for ordinal data
 - Suitable for high-cardinality features
 - Preserves uniqueness of categories

ENCODING CATEGORICAL VARIABLES

```
]:\nle = LabelEncoder()\ncategorical_cols = df.select_dtypes(include="object").columns\n\nfor col in categorical_cols:\n    df[col] = le.fit_transform(df[col].astype(str))\n\nprint(df.info())
```

5.3 BEFORE VS AFTER COMPARISON WITH VISUALS



- Before scaling, features have different ranges and distributions. After scaling, all features are standardized with mean around 0 and similar spread, improving model performance.

6.ADVANCED ANALYSIS

6.1 CORRELATION ANALYSIS

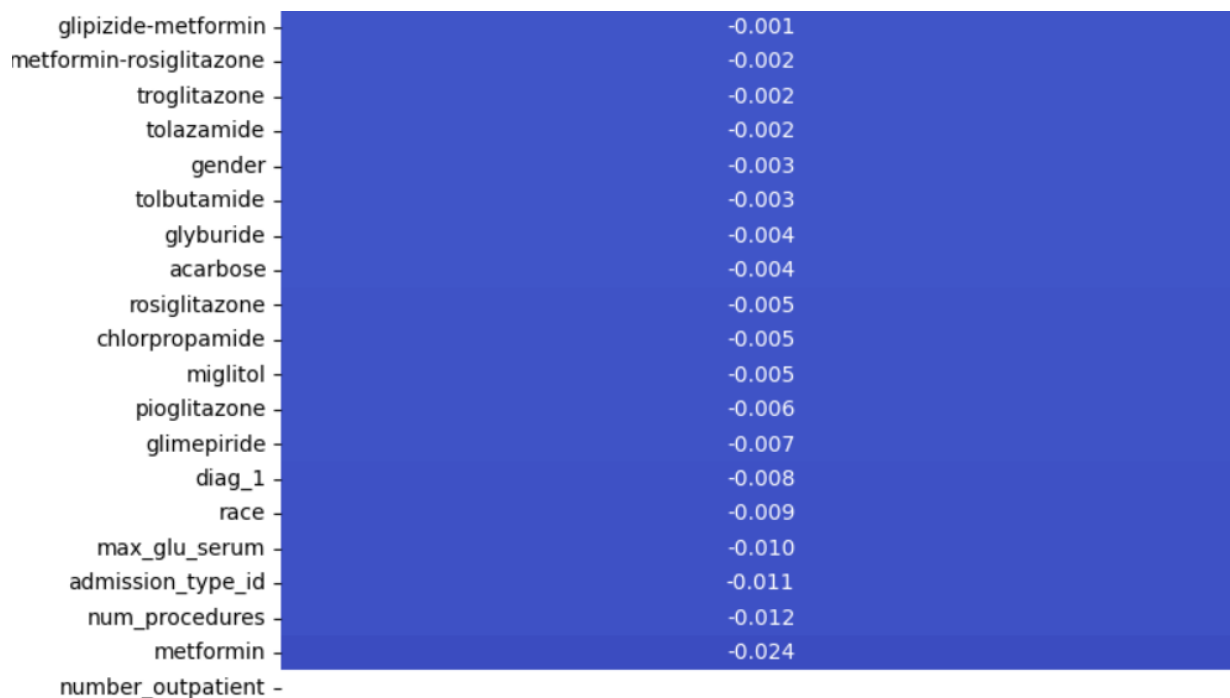
Correlation analysis is used to measure the strength and direction of relationships between numerical variables in the dataset.

It helps in understanding how features interact with each other and whether any strong dependencies exist.

In readmission analysis, correlation analysis is particularly useful for identifying whether any single feature has a direct influence on readmission or whether readmission is driven by a combination of multiple factors.

Correlation of Features with readmit_30

readmit_30	1.000
number_inpatient	0.148
total_visits	0.148
number_diagnoses	0.050
discharge_disposition_id	0.049
time_in_hospital	0.045
num_medications	0.042
long_stay	0.030
diabetesMed	0.027
num_lab_procedures	0.020
change	0.020
age	0.018
A1Cresult	0.017
repaglinide	0.010
medical_specialty	0.010
admission_source_id	0.006
diag_3	0.003
insulin	0.003
diag_2	0.001
glipizide	0.000
nateglinide	0.000
glyburide-metformin	-0.001
acetoexamide	-0.001
metformin-pioglitazone	-0.001
glimepiride-pioglitazone	-0.001



The correlation heatmap shows the relationship between different features and the target variable *readmit_30*. Overall, most features have very low correlation values (close to 0), indicating that no single variable strongly influences readmission on its own. However, a few features show relatively higher positive correlations.

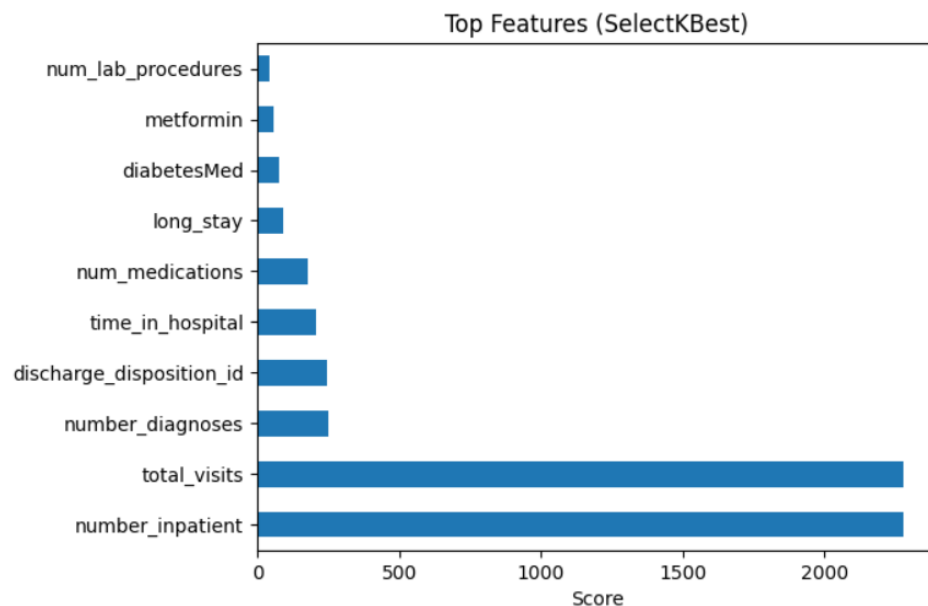
In particular, *number_inpatient* and *total_visits* have the highest correlation (around 0.148), suggesting that patients with more previous hospital visits are more likely to be readmitted. Other features such as *number_diagnoses*, *time_in_hospital*, and *num_medications* show mild positive relationships, indicating that treatment complexity and longer hospital stays slightly increase readmission risk.

On the other hand, several features show very weak or slightly negative correlations, such as *number_outpatient*, *metformin*, and some medication-related variables. These values are very close to zero, meaning they have minimal impact on predicting readmission. Demographic features like *age*, *gender*, and *race* also show very weak correlations, suggesting that clinical history and hospital utilization are more important factors than demographics.

In summary, the graph indicates that readmission is influenced by a combination of multiple small factors rather than a single dominant feature, with prior hospital visits and treatment intensity being the most relevant contributors.

6.2 FEATURE SELECTION USING SELECTKBEST

Feature selection is a critical step in the analytical workflow that focuses on identifying the most relevant features contributing to readmission analysis. In large patient datasets, not all variables provide meaningful information, and some may introduce redundancy or noise. Therefore, selecting important features helps in improving interpretability, reducing dimensional complexity, and enhancing the overall quality of analysis.

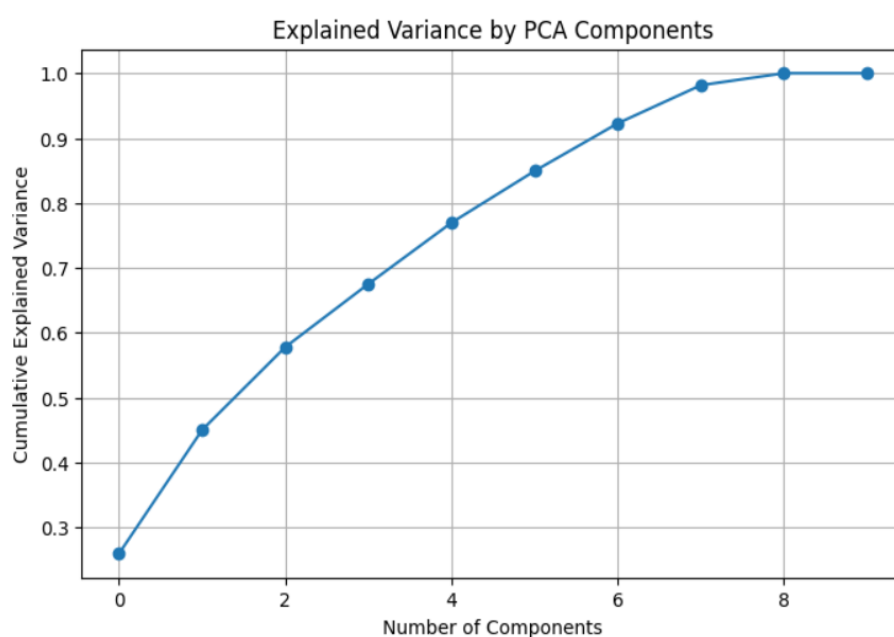


The SelectKBest feature importance graph shows that *number_inpatient* and *total_visits* are the most influential factors in predicting readmission, with much higher scores than other features. This indicates that patients with frequent hospital visits, especially inpatient visits, are more likely to be readmitted.

Features like *number_diagnoses*, *discharge_disposition_id*, and *time_in_hospital* have moderate importance, suggesting that disease complexity and hospital stay duration also affect readmission. In contrast, variables such as *diabetesMed*, and *num_lab_procedures* have lower impact. Overall, past hospital usage and treatment complexity are the key factors influencing readmission risk.

6.3 PRINCIPAL COMPONENT ANALYSIS

Principal Component Analysis was applied to reduce the dimensionality of the dataset while preserving maximum variance. This technique transforms the original features into a new set of orthogonal components that capture the most important information.



PCA WITH 95% VARIANCE

The PCA explained variance graph shows how much information (variance) is captured as the number of principal components increases. Initially, the first few components contribute significantly, with around 45% variance explained by the first two components and nearly 75% by the first four. As more components are added, the increase in explained variance gradually slows down. Around 6 to 7 components, the model captures over 90–95% of the total variance, indicating that most of the important information in the dataset can be represented using fewer dimensions. After this point, adding more components provides very little additional value. This suggests that reducing the dataset to about 6–7 principal components is sufficient for effective analysis while minimizing complexity.

7.VISUALISATION AND DASHBOARD

7.1 Tools Used

The project utilized a combination of modern data visualization and dashboarding tools to effectively present insights:

- Power BI: Used for building the interactive dashboard, enabling dynamic filtering, slicers, and real-time visual interaction.
- Python (Matplotlib & Seaborn): Used for initial exploratory data analysis and static visualizations.
- Plotly: Used for creating interactive plots that enhance user engagement and deeper data exploration.
- Jupyter Notebook: Served as the development environment for analysis, preprocessing, and visualization.

These tools together helped in transforming raw healthcare data into meaningful and interactive visual insights.



7.2 Types of Visualizations Used

A variety of visualizations were used to analyze and present the data effectively:

- **Bar Charts:**
Used to compare readmission status with features like emergency visits, medications, and total visits.
- **Line Chart:**
Used to show trends such as average hospital stay across readmission categories.
- **Scatter Plot:**
Displays the relationship between *time in hospital* and *number of medications*, highlighting readmission patterns.
- **Donut Chart (Pie Chart):**
Represents the distribution of long-stay vs short-stay patients.

7.3 Dashboard Explanation

Used for comparing average values like total visits across readmission groups. The dashboard provides an interactive overview of diabetic patient readmission patterns using filters for gender, age, and race. It highlights that patients with longer hospital stays, higher medication counts, and more emergency or total visits are more likely to be readmitted. The visualizations also show that while most patients are not readmitted, a notable portion remains at risk. Overall, the dashboard combines multiple insights in a clear and user-friendly way, helping to identify high-risk patients and support better healthcare decisions.

8.INSIGHTS AND STORY TELLING

8.1 Key Findings from Analysis

The analysis revealed that hospital readmission is primarily influenced by factors related to patient medical history and healthcare utilization.

Variables such as number of inpatient visits, emergency visits, and total visits showed the strongest impact on readmission. Patients with longer hospital stays, higher number of diagnoses, and increased medication usage were also more likely to be readmitted, indicating greater treatment complexity.

Additionally, feature selection and correlation analysis confirmed that clinical and hospital-related factors have more influence than demographic attributes like age, gender, and race.

8.2 Patterns and Trends Identified

Several meaningful patterns were observed during the analysis. Most patients were discharged within a short duration (around 3–5 days), but patients with longer hospital stays showed a higher probability of readmission.

A clear trend was seen where patients with frequent past hospital visits, especially inpatient visits, had a consistently higher risk. The data also indicated that as the number of diagnoses and medications increases, the likelihood of readmission also increases. Overall, the trend suggests that readmission is a result of cumulative health and treatment factors rather than a single cause.

8.3 Business / Social Insights

From a business perspective, hospital readmissions lead to increased operational costs, resource utilization, and workload for healthcare providers. By identifying high-risk patients early, hospitals can optimize resource allocation, reduce unnecessary admissions, and improve service efficiency. Socially, effective management of diabetic patients can lead to better health outcomes, reduced complications, and improved quality of life. This study highlights how data-driven insights can bridge the gap between clinical data and practical

healthcare improvements.

8.4 Decision-Making Based on Results

The results of this analysis support strategic and data-driven decision-making in healthcare systems. Hospitals can use these insights to prioritize patients with high hospital visit history, longer stays, and complex treatment profiles for additional care and monitoring. Implementing targeted discharge planning, regular follow-ups, and personalized care strategies can significantly reduce readmission rates. Furthermore, healthcare administrators can use these findings to design policies and allocate resources more effectively, ensuring better patient outcomes while maintaining operational efficiency.

9.CONCLUSION

9.1 Summary of the Project

This project analyzed hospital readmission patterns among diabetic patients using a structured data science approach. It began with data cleaning and preprocessing to handle missing values, inconsistencies, and outliers. Exploratory Data Analysis (EDA) was then performed to understand distributions, relationships, and trends in the data. Further, feature engineering and transformation techniques were applied to prepare the dataset for advanced analysis. Correlation analysis and PCA helped identify the most influential factors and reduce dimensionality. Finally, visualizations and an interactive dashboard were developed to clearly communicate insights. Overall, the project demonstrated how raw healthcare data can be converted into meaningful information for better understanding and decision-making.

9.2 Outcomes Achieved

The project achieved several important outcomes. It successfully identified key factors influencing readmission, such as number of inpatient visits, emergency visits, hospital stay duration, number of diagnoses, and medication count.

Data quality was significantly improved through preprocessing steps, making the dataset more reliable for analysis. Various visualizations helped uncover hidden patterns and relationships within the data. The development of an interactive dashboard enhanced the presentation of insights and allowed easy interpretation of complex information. Additionally, the project highlighted the importance of healthcare data analysis in improving patient care and reducing hospital burden.

9.3 Limitations

- Presence of missing, inconsistent, and unknown values in several features
- Lack of detailed patient-level information such as lifestyle, diet, and medical history outside hospital records
- Dataset limited to a specific time period and hospitals, which may affect generalization
- Readmission influenced by external factors not captured in the dataset
- No implementation of predictive or real-time machine learning models
- Possible class imbalance affecting analysis outcomes

9.4 Future Scope

- Develop and evaluate machine learning models for accurate readmission prediction
- Incorporate real-time data integration from hospital systems
- Include additional features such as lifestyle, behavioral, and environmental factors
- Apply advanced techniques like deep learning for improved analysis
- Enhance dashboard with real-time monitoring and predictive insights
- Implement automated alert systems for identifying high-risk patients early
- Extend the study to other diseases and broader healthcare datasets.

10. REFERENCES

10.1 Dataset Sources

Dataset Source: Kaggle – Diabetes 130-US Hospitals Dataset

Dataset Link: <https://www.kaggle.com/datasets/brandao/diabetes>

10.2 Tools and Libraries Used

- **Pandas** – Used for data loading, cleaning, and manipulation.
- **NumPy** – Used for numerical operations and array computations.
- **Matplotlib** – Used for creating basic data visualizations.
- **Seaborn** – Used for advanced statistical and graphical visualizations.
- **Scikit-learn** – Used for feature selection (SelectKBest) and PCA.

10.3 Research Papers / Websites Referred

- Kaggle Dataset Documentation – Provided detailed information about the dataset structure, features, and context of diabetic patient records.
- Scikit-learn Official Documentation – Used for understanding preprocessing techniques, feature selection methods, and Principal Component Analysis (PCA).
- Matplotlib and Seaborn Documentation – Referenced for creating various statistical and distribution plots during EDA.
- Plotly Documentation – Used for creating interactive visualizations.
- Streamlit / Power BI Documentation – Helped in building and designing the interactive dashboard.

Research Articles and Academic References

- Studies on hospital readmission prediction and risk factors in diabetic patients
- Research papers on healthcare analytics and data-driven decision-making.
- Articles on the impact of hospital stay duration and treatment complexity on patient outcomes
- Literature on dimensionality reduction techniques like PCA in healthcare Datasets.

Web Resources

- Towards Data Science (Medium) – Articles related to EDA, feature engineering, and healthcare analytics
- GeeksforGeeks – Reference for Python concepts, preprocessing techniques, and algorithms
- Stack Overflow – Used for resolving coding errors and implementation issues.

Appendix

GitHub Link

<https://github.com/guru16-02/TEAM-17-24ADI204.git>

ONEDRIVE LINK:

https://kumaragurudtsteam-my.sharepoint.com/:f:/g/personal/anusri_24ad_kct_ac_in/IgDNfJyctDzZTox4MlIck6uqAR5CLjtEoHjoRbHU9x1dry4?e=cJE3R4

CODE SNIPPETS

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
from sklearn.feature_selection import SelectKBest, f_classif
```

LOAD THE DATASET

```
df=pd.read_csv(r'C:\Users\Welcome\Downloads\archive (1) (1).csv')
```

BASIC OPERATION TO UNDERSTAND THE DATASET

```
df.head()
```

	encounter_id	patient_nbr	race	gender	age	weight	admission_type_id	discharge_disposition_id	admission_source_id	time_in_hospital	...	c
0	2278392	8222157	Caucasian	Female	[0-10)	?	6	25	1	1	...	
1	149190	55629189	Caucasian	Female	[10-20)	?	1	1	7	3	...	
2	64410	86047875	AfricanAmerican	Female	[20-30)	?	1	1	7	2	...	
3	500364	82442376	Caucasian	Male	[30-40)	?	1	1	7	2	...	
4	16680	42519267	Caucasian	Male	[40-50)	?	1	1	7	1	...	

5 rows × 50 columns

```
df.shape
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 101766 entries, 0 to 101765
```

```
Data columns (total 50 columns):
```

#	Column	Non-Null Count	Dtype
0	encounter_id	101766 non-null	int64
1	patient_nbr	101766 non-null	int64
2	race	101766 non-null	object
3	gender	101766 non-null	object
4	age	101766 non-null	object
5	weight	101766 non-null	object
6	admission_type_id	101766 non-null	int64
7	discharge_disposition_id	101766 non-null	int64
8	admission_source_id	101766 non-null	int64
9	time_in_hospital	101766 non-null	int64
10	payer_code	101766 non-null	object
11	medical_specialty	101766 non-null	object
12	num_lab_procedures	101766 non-null	int64
13	num_procedures	101766 non-null	int64
14	num_medications	101766 non-null	int64
15	number_outpatient	101766 non-null	int64
16	number_emergency	101766 non-null	int64
17	number_inpatient	101766 non-null	int64
18	diag_1	101766 non-null	object
19	diag_2	101766 non-null	object
20	diag_3	101766 non-null	object
21	number_diagnoses	101766 non-null	int64
22	max_glu_serum	5346 non-null	object
23	A1Cresult	17018 non-null	object
24	metformin	101766 non-null	object
25	repaglinide	101766 non-null	object
26	nateglinide	101766 non-null	object
27	chlorpropamide	101766 non-null	object
28	glimepiride	101766 non-null	object
29	glipizide	101766 non-null	object

```

29  acetoexamide          101766 non-null object
30  glipizide             101766 non-null object
31  glyburide             101766 non-null object
32  tolbutamide           101766 non-null object
33  pioglitazone          101766 non-null object
34  rosiglitazone         101766 non-null object
35  acarbose              101766 non-null object
36  miglitol              101766 non-null object
37  troglitazone          101766 non-null object
38  tolazamide            101766 non-null object
39  examide               101766 non-null object
40  citoglipton           101766 non-null object
41  insulin               101766 non-null object
42  glyburide-metformin   101766 non-null object
43  glipizide-metformin   101766 non-null object
44  glimepiride-pioglitazone 101766 non-null object
45  metformin-rosiglitazone 101766 non-null object
46  metformin-pioglitazone 101766 non-null object
47  change                101766 non-null object
48  diabetesMed           101766 non-null object
49  readmitted            101766 non-null object

```

dtypes: int64(13), object(37)

memory usage: 38.8+ MB

```
df.describe()
```

encounter_id	patient_nbr	admission_type_id	discharge_disposition_id	admission_source_id	time_in_hospital	num_lab_procedures	num_procedures	num_
1.017660e+05	1.017660e+05	101766.000000	101766.000000	101766.000000	101766.000000	101766.000000	101766.000000	10
1.652016e+08	5.433040e+07	2.024006	3.715642	5.754437	4.395987	43.095641	1.339730	
1.026403e+08	3.869636e+07	1.445403	5.280166	4.064081	2.985108	19.674362	1.705807	
1.252200e+04	1.350000e+02	1.000000	1.000000	1.000000	1.000000	1.000000	0.000000	
8.496119e+07	2.341322e+07	1.000000	1.000000	1.000000	2.000000	31.000000	0.000000	
1.523890e+08	4.550514e+07	1.000000	1.000000	7.000000	4.000000	44.000000	1.000000	
2.302709e+08	8.754595e+07	3.000000	4.000000	7.000000	6.000000	57.000000	2.000000	
4.438672e+08	1.895026e+08	8.000000	28.000000	25.000000	14.000000	132.000000	6.000000	

```
df.isnull().sum()
```

encounter_id	0
patient_nbr	0
race	0
gender	0
age	0
weight	0
admission_type_id	0
discharge_disposition_id	0
admission_source_id	0
time_in_hospital	0
payer_code	0
medical_specialty	0
num_lab_procedures	0
num_procedures	0
num_medications	0
number_outpatient	0
number_emergency	0
number_inpatient	0
diag_1	0
diag_2	0
diag_3	0
number_diagnoses	0
max_glu_serum	96420
A1Cresult	84748
metformin	0
repaglinide	0
nateglinide	0
chlorpropamide	0
glimepiride	0
acetohexamide	0
glipizide	0
glyburide	0
tolbutamide	0
pioglitazone	0

```

rosiglitazone      0
acarbose           0
miglitol           0
troglitazone       0
tolazamide         0
examide            0
citoglipton        0
insulin            0
glyburide-metformin 0
glipizide-metformin 0
glimepiride-pioglitazone 0
metformin-rosiglitazone 0
metformin-pioglitazone 0
change            0
diabetesMed        0
readmitted         0
dtype: int64

```

HANDLE THE MISSING AND CATAGERICAL FEATURES

```

: df.replace('?',np.nan, inplace=True)

```

```

: df.drop(columns=["weight", "payer_code"], inplace=True)

```

```

: print(df.race)

```

```

0          Caucasian
1          Caucasian
2    AfricanAmerican
3          Caucasian
4          Caucasian
...
101761    AfricanAmerican
101762    AfricanAmerican
101763          Caucasian
101764          Caucasian
101765          Caucasian
Name: race, Length: 101766, dtype: object

```

```

: df["race"].isnull()

```

```

: 0          False
  1          False
  2          False
  3          False
  4          False
  ...
101761      False
101762      False
101763      False

```

```
101760    False
...
101761    False
101762    False
101763    False
101764    False
101765    False
Name: race, Length: 101766, dtype: bool
```

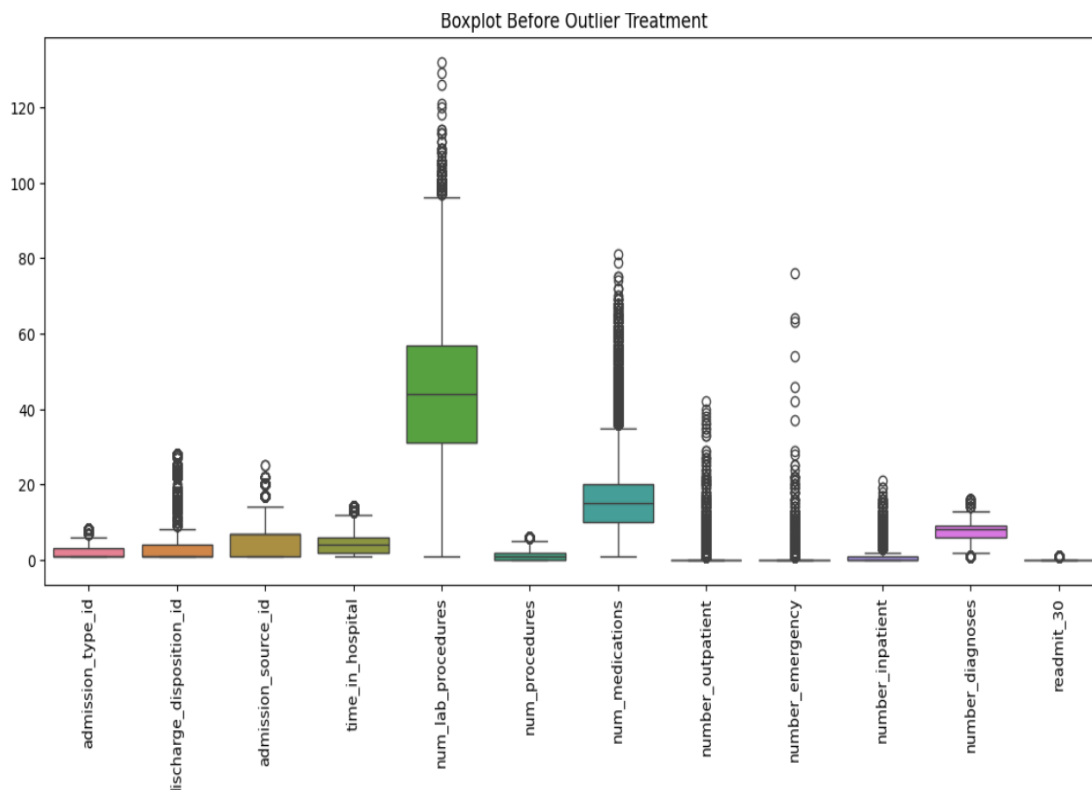
```
df["race"].fillna("Unknown", inplace=True)
df["medical_specialty"].fillna("Unknown", inplace=True)
```

HANDLE THE MISSING AND CATAGERICAL FEATURES

-----WEEK - 3-----

```
df.replace('?',np.nan, inplace=True)
```

	encounter_id	patient_nbr	race	gender	age	weight	admission_type_id	discharge_disposition_id	admission_source_id	time_in_hospital
0	2278392	8222157	Caucasian	Female	[0-10)	NaN	6	25	1	1
1	149190	55629189	Caucasian	Female	[10-20)	NaN	1	1	7	3
2	64410	86047875	AfricanAmerican	Female	[20-30)	NaN	1	1	7	2
3	500364	82442376	Caucasian	Male	[30-40)	NaN	1	1	7	2
4	16680	42519267	Caucasian	Male	[40-50)	NaN	1	1	7	1
...	...	...	...	...	...	...	...	...	...	...
101761	443847548	100162476	AfricanAmerican	Male	[70-80)	NaN	1	3	7	3
101762	443847782	74694222	AfricanAmerican	Female	[80-90)	NaN	1	4	5	5
101763	443854148	41088789	Caucasian	Male	[70-80)	NaN	1	1	7	1



```
2]: num_cols = df.select_dtypes(include=["int64", "float64"]).columns
num_cols

2]: Index(['admission_type_id', 'discharge_disposition_id', 'admission_source_id',
        'time_in_hospital', 'num_lab_procedures', 'num_procedures',
        'num_medications', 'number_outpatient', 'number_emergency',
        'number_inpatient', 'number_diagnoses', 'readmit_30'],
        dtype='object')
```

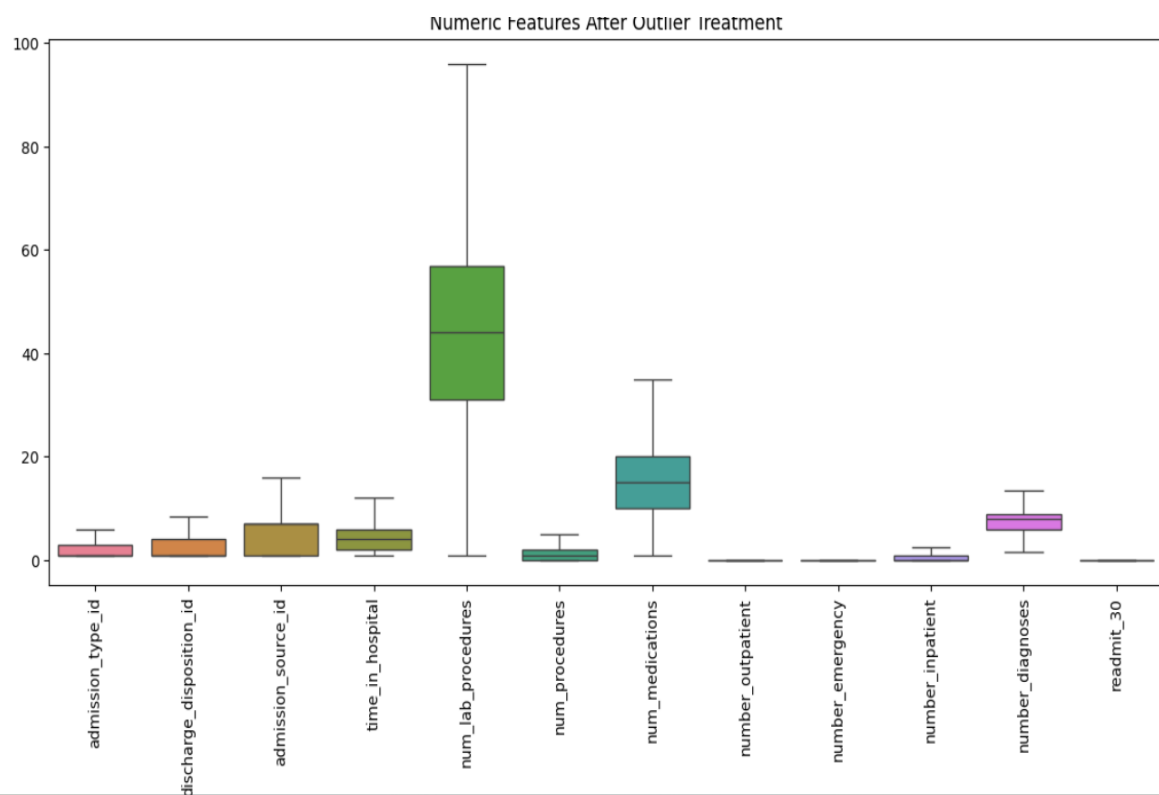
HANDLING HE OUTLIER USING THE IQR

```
3]: for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    df[col] = df[col].clip(lower, upper)

4]: plt.figure(figsize=(14,6))
sns.boxplot(data=df[num_cols])
plt.xticks(rotation=90)
plt.title("Numeric Features After Outlier Treatment")
plt.show()
```



```
df["gender"] = df["gender"].map({"Male": 1, "Female": 0})
df["diabetesMed"] = df["diabetesMed"].map({"Yes": 1, "No": 0})
df["insulin"] = df["insulin"].map({"Yes": 1, "No": 0})
```

```
categorical_cols = [
    "admission_type_id",
    "discharge_disposition_id",
    "admission_source_id",
    "A1Cresult"
]

df = pd.get_dummies(df, columns=categorical_cols, drop_first=True)
```

```
print("Final dataset shape:", df.shape)
print("Total missing values:", df.isna().sum().sum())
print(df.head())
```

Final dataset shape: (101766, 71)

Total missing values: 152608

	race	gender	age	time_in_hospital	\
0	Caucasian	0.0	[0-10)	1	
1	Caucasian	0.0	[10-20)	3	
2	AfricanAmerican	0.0	[20-30)	2	
3	Caucasian	1.0	[30-40)	2	
4	Caucasian	1.0	[40-50)	1	

	medical_specialty	num_lab_procedures	num_procedures	\
0	Pediatrics-Endocrinology	41	0	
1	Unknown	59	0	
2	Unknown	11	5	
3	Unknown	44	1	
4	Unknown	51	0	

	num_medications	number_outpatient	number_emergency	...	\
0	1	0	0	...	
1	18	0	0	...	
2	13	0	0	...	
3	16	0	0	...	
4	8	0	0	...	

	admission_source_id_7	admission_source_id_8	admission_source_id_9	\
0	False	False	False	
1	True	False	False	
2	True	False	False	
3	True	False	False	
4	True	False	False	

	admission_source_id_10	admission_source_id_11	admission_source_id_13	\
0	False	False	False	
1	False	False	False	
2	False	False	False	
3	False	False	False	
4	False	False	False	

	admission_source_id_14	admission_source_id_16	A1Cresult_>8	A1Cresult_Norm
0	False	False	False	False
1	False	False	False	False
2	False	False	False	False
3	False	False	False	False
4	False	False	False	False

[5 rows x 71 columns]

```
num_cols = df.select_dtypes(include=["int64", "float64"]).columns
num_cols
```

```
Index(['gender', 'time_in_hospital', 'num_lab_procedures', 'num_procedures',
      'num_medications', 'number_outpatient', 'number_emergency',
      'number_inpatient', 'number_diagnoses', 'insulin', 'diabetesMed',
      'readmit_30'],
      dtype='object')
```

```
from scipy.stats import zscore
z_scores = df[num_cols].apply(zscore)
z_scores.head()
```

	gender	time_in_hospital	num_lab_procedures	num_procedures	num_medications	number_outpatient	number_emergency	number_inpatient	number_di
0	NaN	-1.163010	-0.106226	-0.816151	-2.002068	NaN	NaN	-0.633137	-3
1	NaN	-0.471487	0.809860	-0.816151	0.296283	NaN	NaN	-0.633137	0
2	NaN	-0.817248	-1.633035	2.344652	-0.379703	NaN	NaN	0.599406	-0
3	NaN	-0.817248	0.046455	-0.183990	0.025889	NaN	NaN	-0.633137	-0
4	NaN	-1.163010	0.402711	-0.816151	-1.055688	NaN	NaN	-0.633137	-1

```
outliers = (z_scores.abs() > 3)
outliers.sum().sort_values(ascending=False)
```

```
number_diagnoses    281
gender              0
num_lab_procedures  0
time_in_hospital    0
num_procedures      0
num_medications     0
number_emergency    0
number_outpatient   0
number_inpatient    0
insulin             0
diabetesMed         0
readmit_30          0
dtype: int64
```

```
outlier_rows = outliers.any(axis=1).sum()
outlier_rows
```

```
np.int64(281)
```

```
Q1 = df['number_diagnoses'].quantile(0.25)
Q3 = df['number_diagnoses'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df['number_diagnoses'] = df['number_diagnoses'].clip(lower, upper)
```

```
print(df.columns)
```

EDA (UNIVARIATE ANALYSIS)

```
plt.figure(figsize=(6,4))
sns.histplot(df['time_in_hospital'], bins=10, kde=True)
plt.title("Distribution of Time in Hospital")
plt.xlabel("Days")
plt.ylabel("Frequency")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['num_medications'])
plt.title("Boxplot of Number of Medications")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.histplot(df['number_inpatient'], bins=10, kde=True)
plt.title("Inpatient Visits Distribution")
plt.xlabel("Visits")
plt.ylabel("Frequency")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.countplot(x=df['readmit_30'])
plt.title("Readmission Distribution")
plt.xlabel("Readmitted (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.countplot(x=df['long_stay'])
plt.title("Long Stay Distribution")
plt.xlabel("Long Stay (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.show()
```

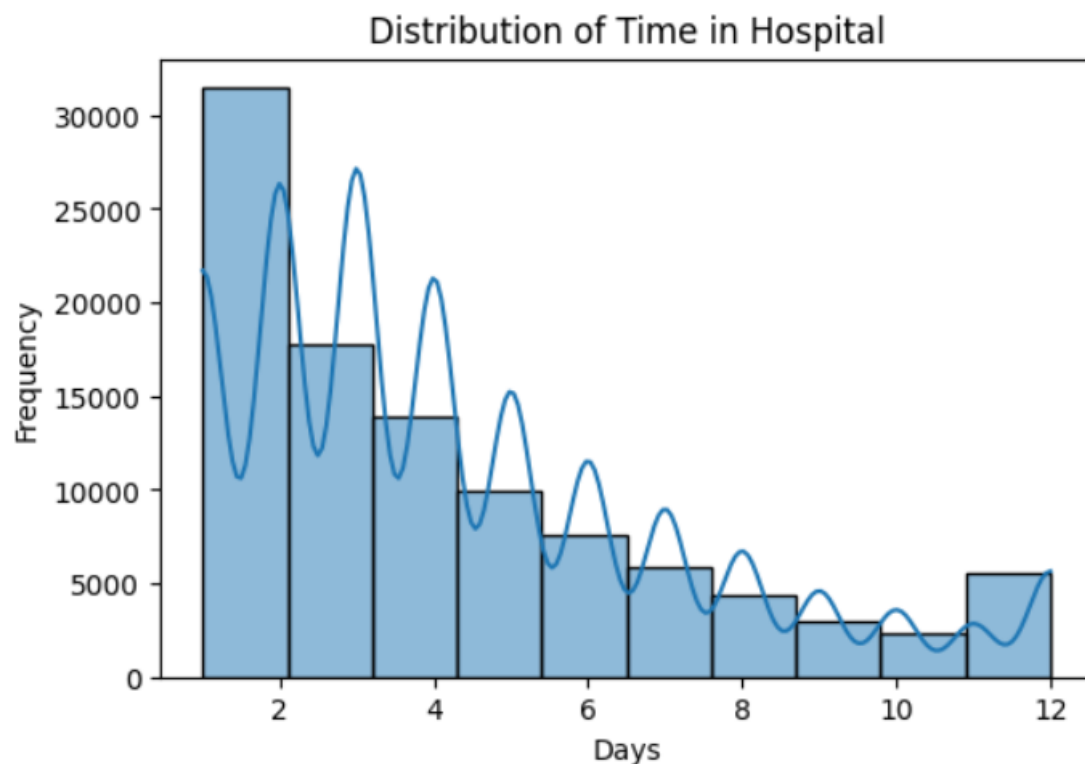
```
df["gender"].value_counts().plot(kind="bar")
plt.title("Gender Distribution")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.histplot(df['number_diagnoses'], bins=10, kde=True)
plt.title("Distribution of Number of Diagnoses")
plt.xlabel("Number of Diagnoses")
plt.ylabel("Frequency")
plt.show()
```

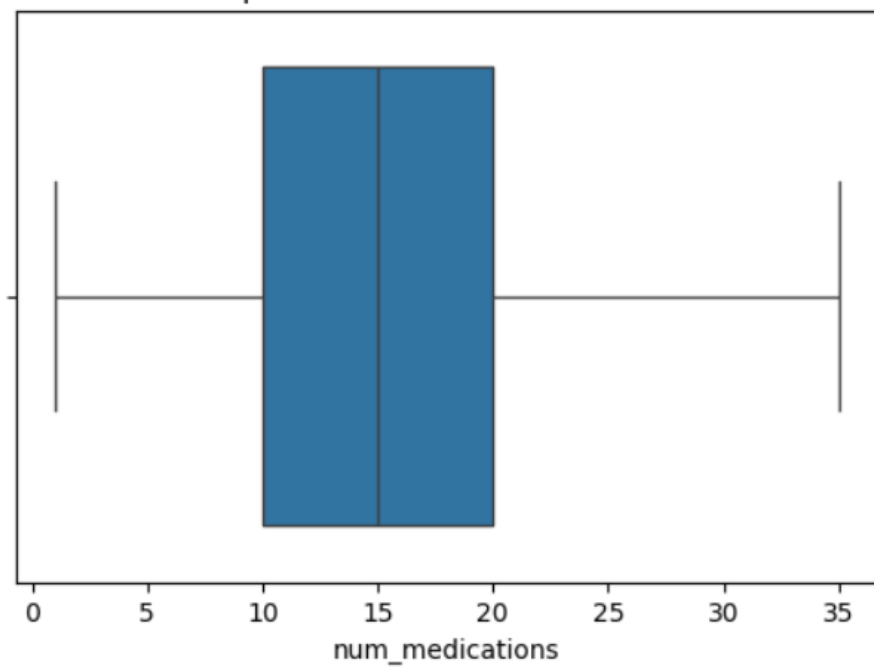
```
plt.figure(figsize=(6,4))
sns.histplot(df['num_lab_procedures'], bins=20, kde=True)
plt.title("Lab Procedures Distribution")
plt.xlabel("Number of Lab Procedures")
plt.ylabel("Frequency")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.countplot(x=df['A1Cresult'])
plt.title("A1C Result Distribution")
plt.show()
```

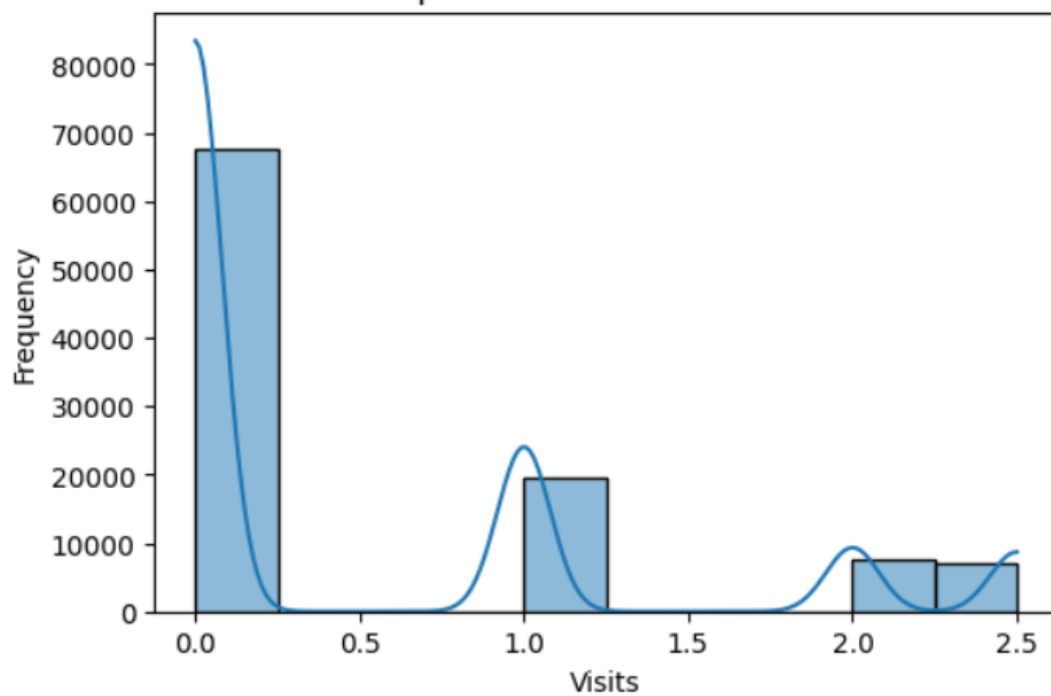
OUTPUT:

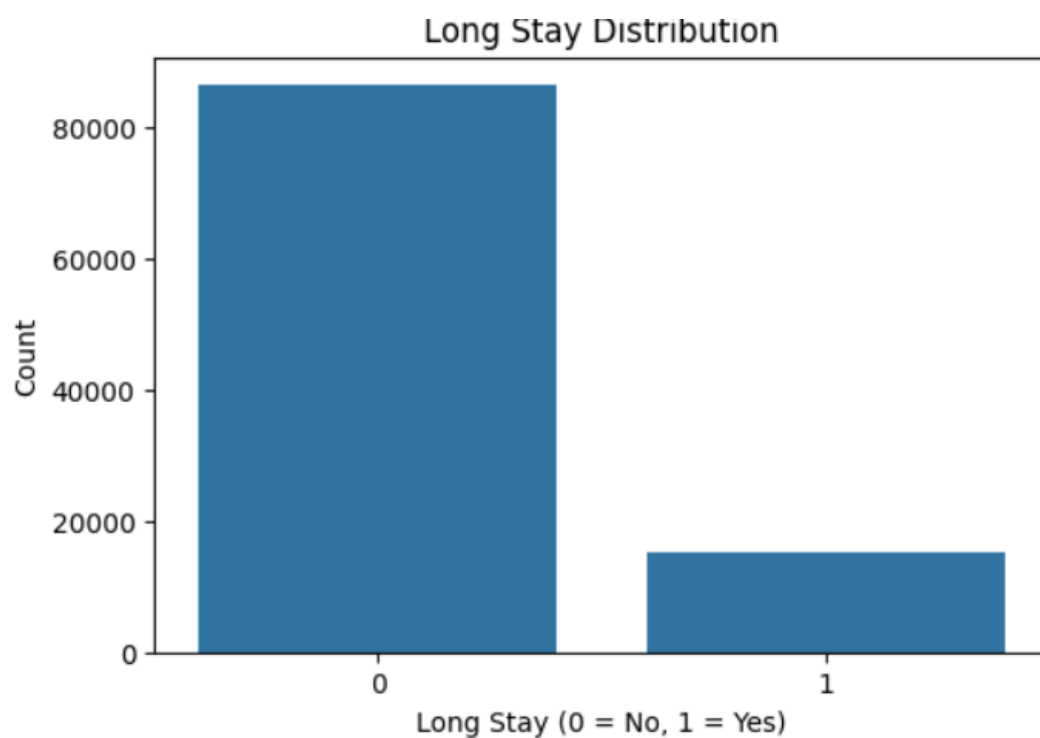
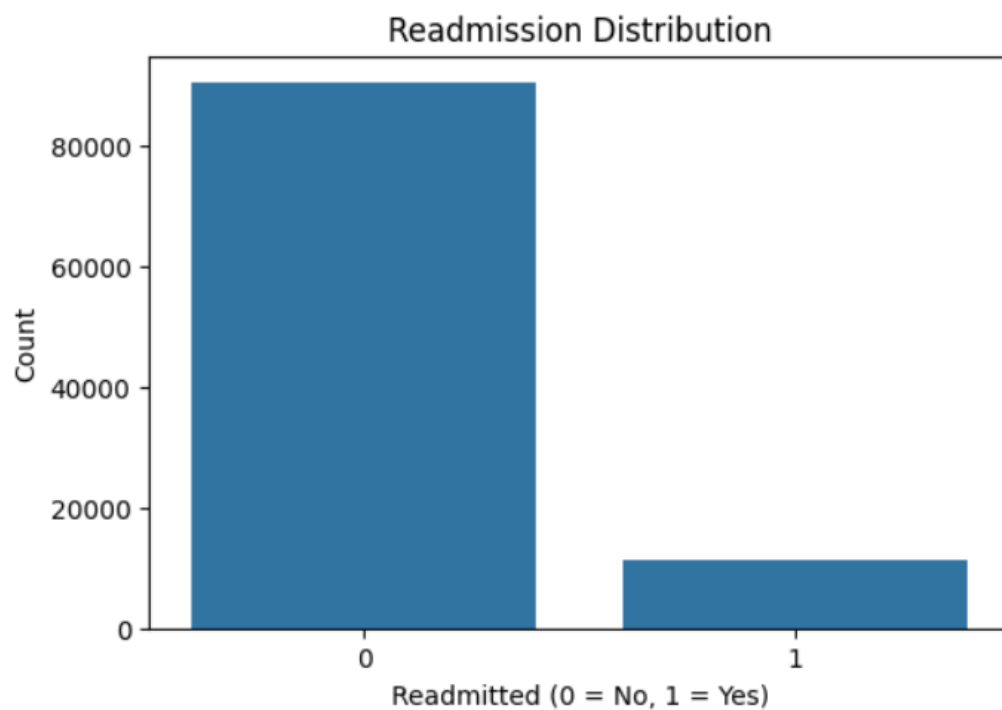


Boxplot of Number of Medications

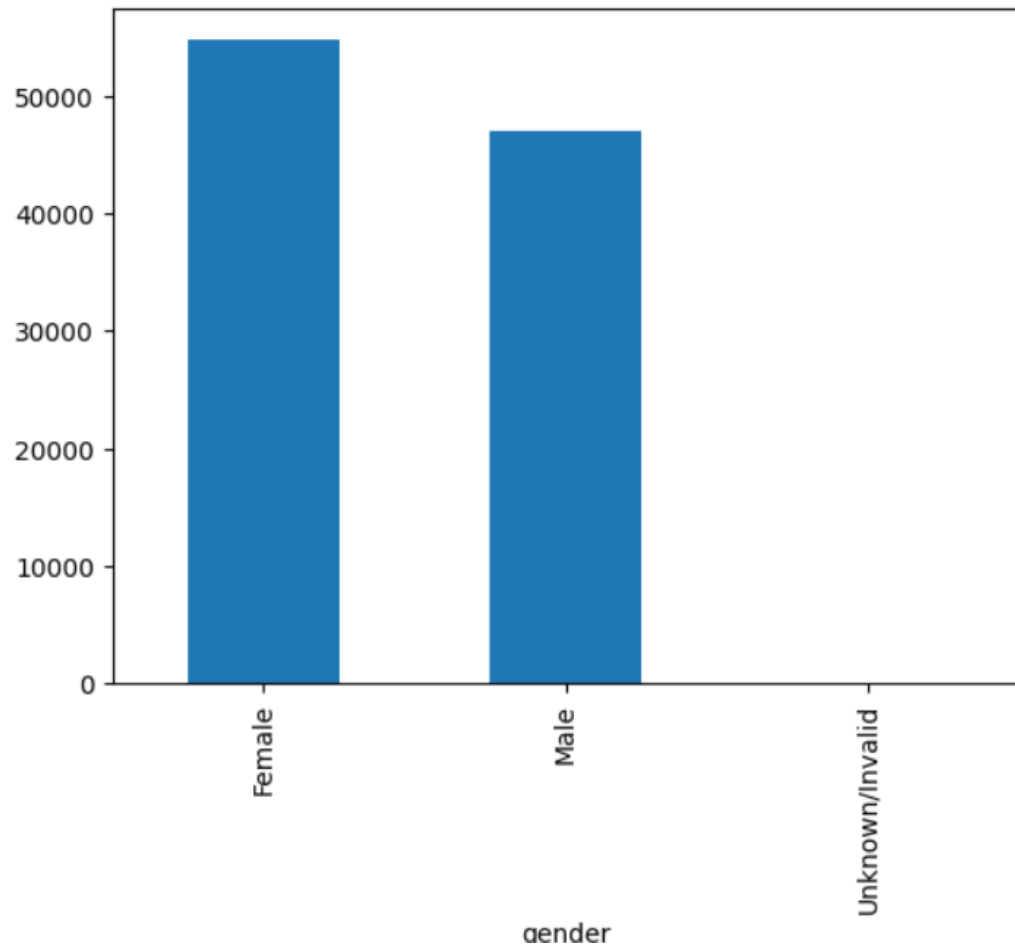


Inpatient Visits Distribution

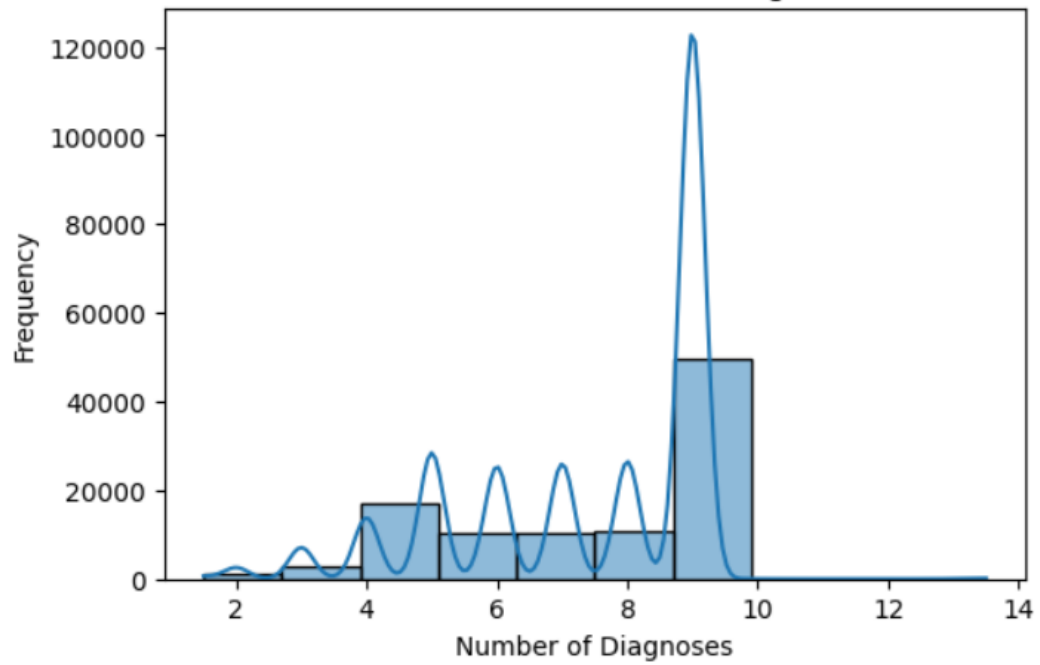




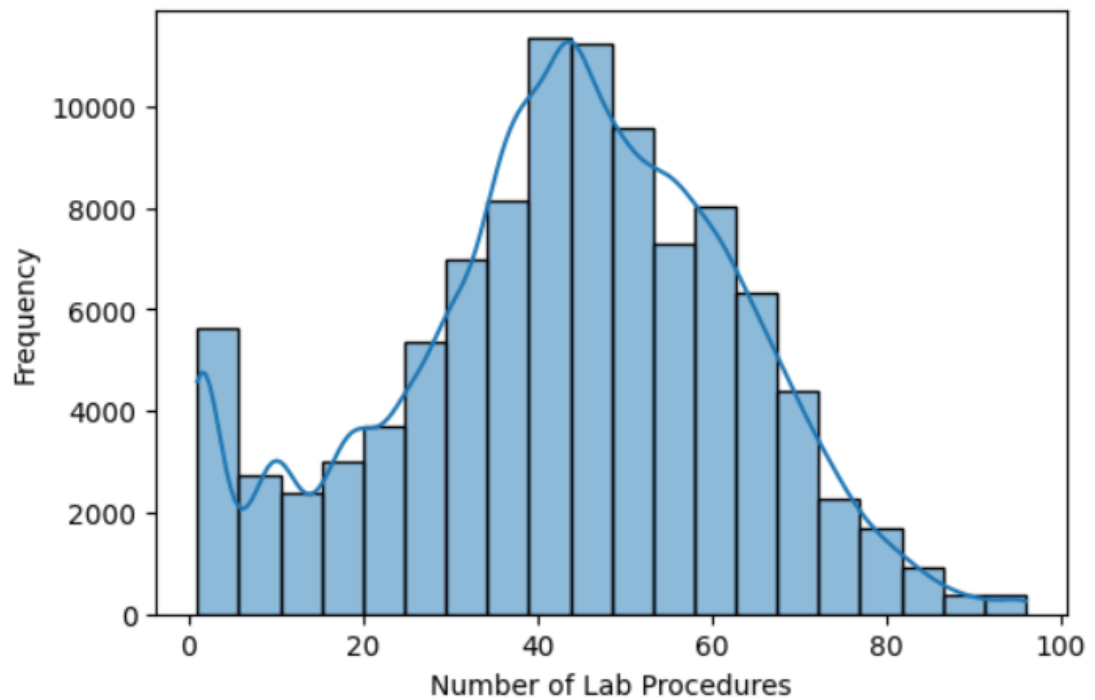
Gender Distribution

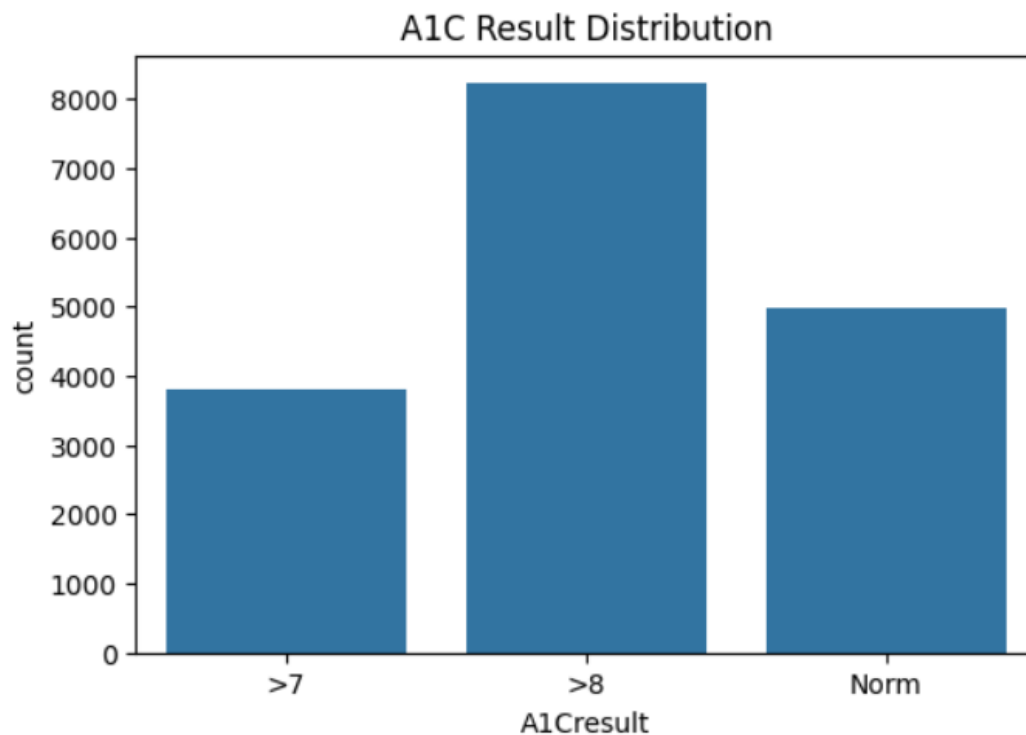


Distribution of Number of Diagnoses



Lab Procedures Distribution





BIVARIATE ANALYSIS:

```
plt.figure(figsize=(6,4))
sns.boxplot(x='readmit_30', y='time_in_hospital', data=df)
plt.title("Time in Hospital vs Readmission")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.barplot(x='readmit_30', y='num_medications', data=df)
plt.title("Medications vs Readmission")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.barplot(x='readmit_30', y='total_visits', data=df)
plt.title("Total Visits vs Readmission")
plt.show()
```

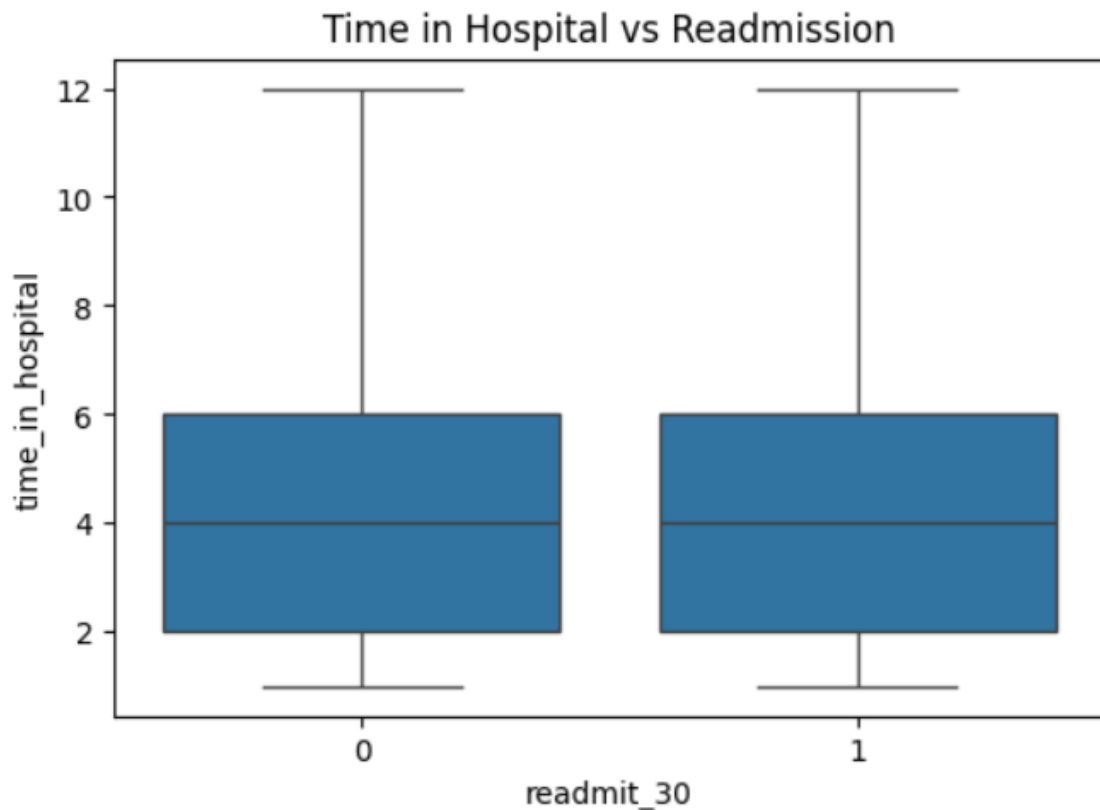
```
plt.figure(figsize=(6,4))
sns.countplot(x='age', hue='readmit_30', data=df)
plt.title("Age vs Readmission")
plt.show()
```

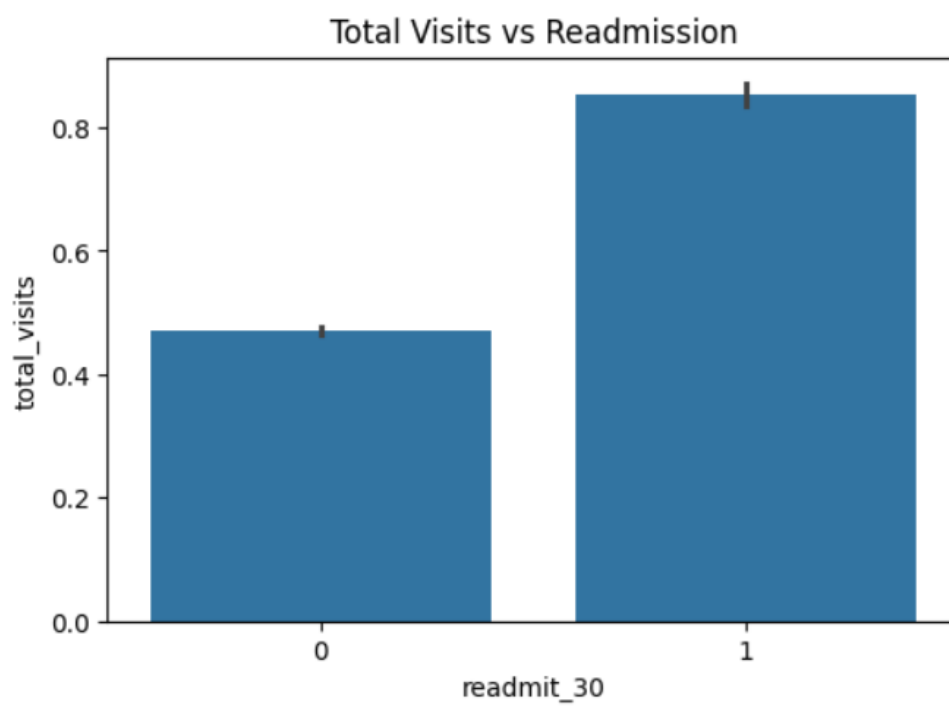
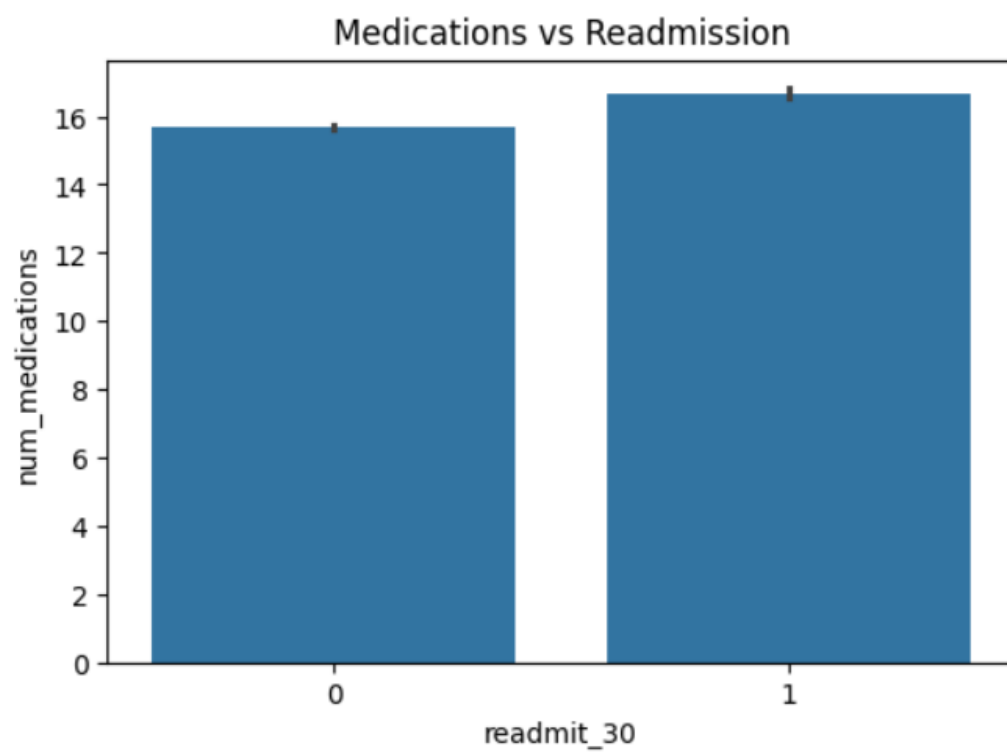
```
plt.figure(figsize=(6,4))
sns.scatterplot(x='time_in_hospital', y='num_medications', hue='readmit_30', data=df)
plt.title("Hospital Stay vs Medications")
plt.show()
```

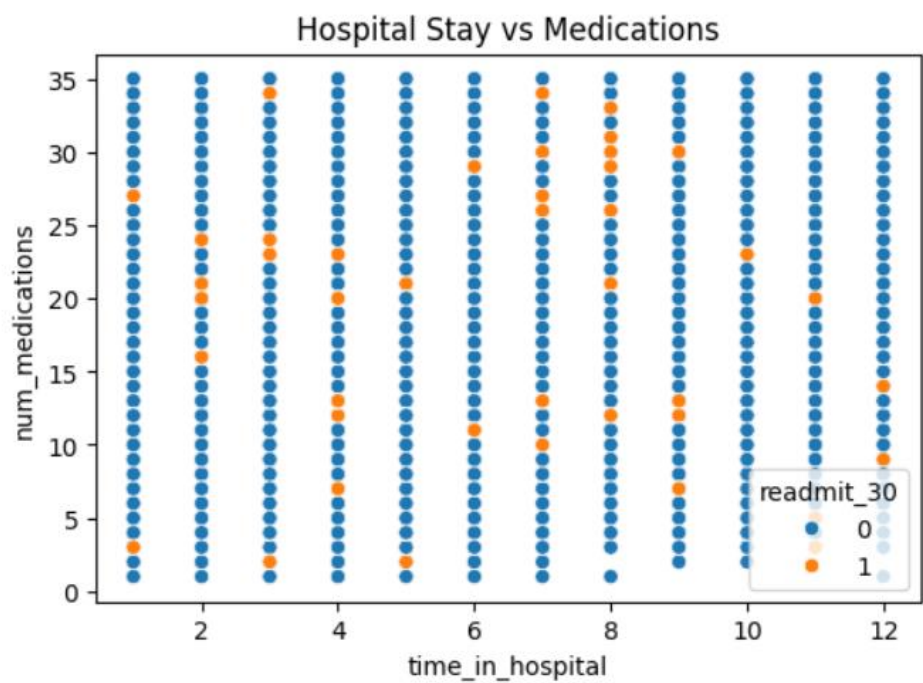
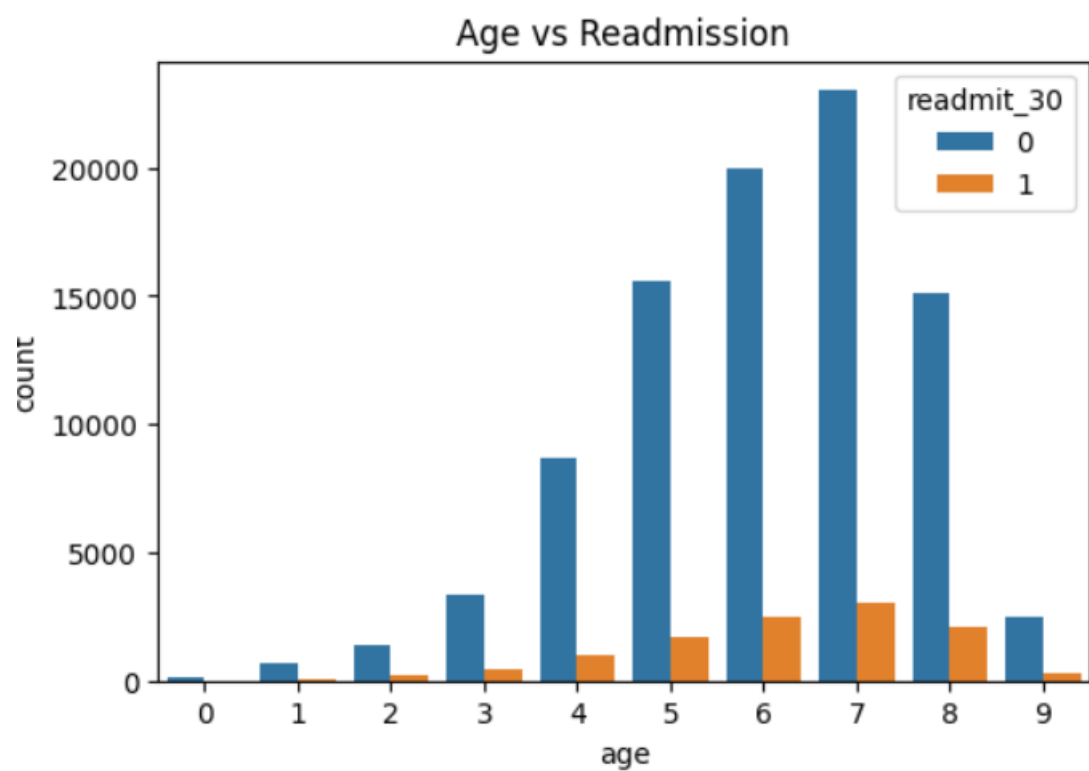
```
plt.figure(figsize=(6,4))
sns.barplot(x='readmit_30', y='number_inpatient', data=df)
plt.title("Inpatient Visits vs Readmission")
plt.show()
```

```
plt.figure(figsize=(6,4))
sns.barplot(x='readmit_30', y='number_diagnoses', data=df)
plt.title("Diagnoses vs Readmission")
plt.show()
```

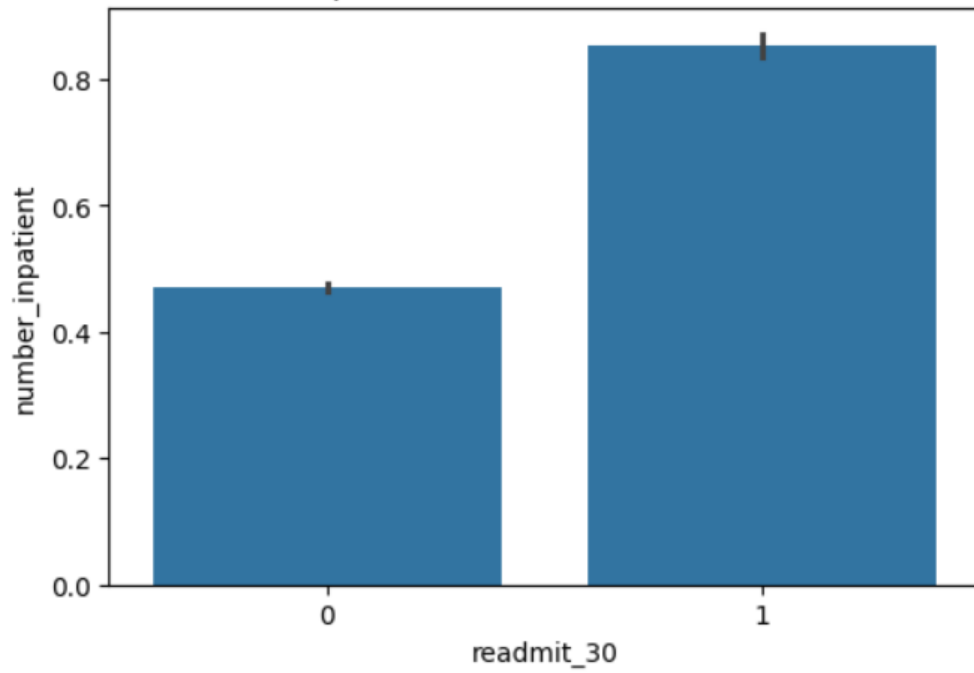
OUTPUT:



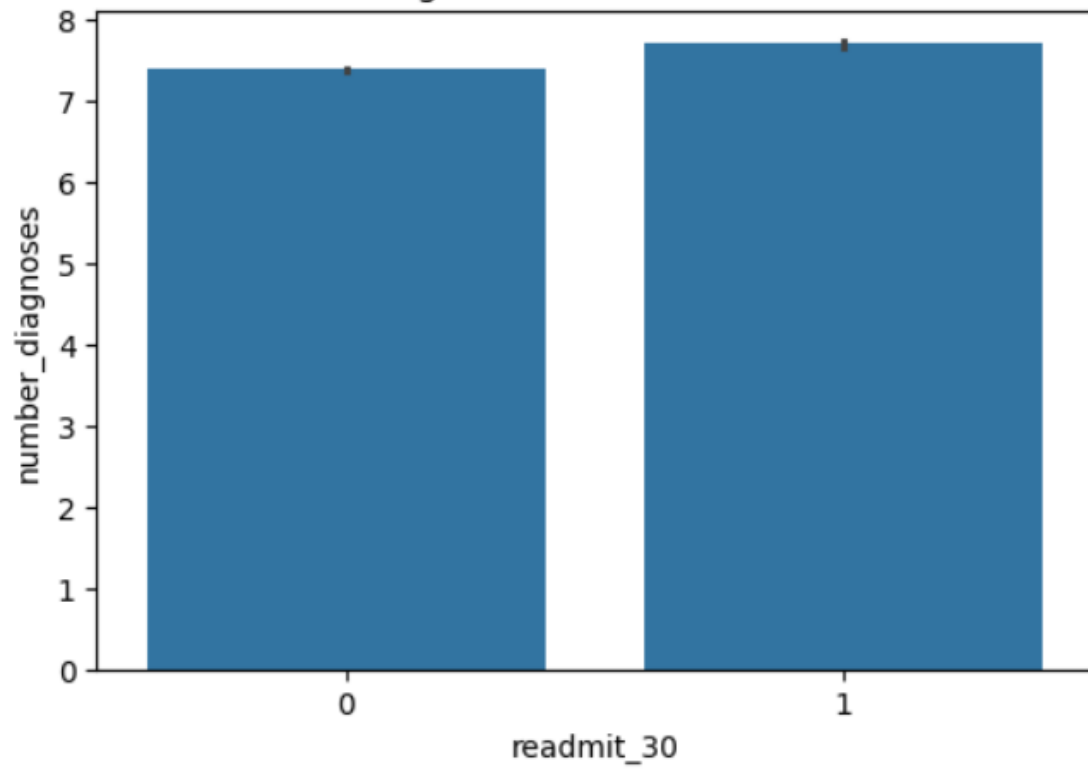




Inpatient Visits vs Readmission



Diagnoses vs Readmission



FEATURE ENGINEERING

```
[47]: df["total_visits"] = (  
        df["number_outpatient"] +  
        df["number_emergency"] +  
        df["number_inpatient"]  
    )  
  
    df["long_stay"] = (df["time_in_hospital"] > 7).astype(int)
```

```
[48]: if "insulin" in df.columns:  
        print(df["insulin"].unique())
```

```
<ArrowStringArray>  
['No', 'Up', 'Steady', 'Down']  
Length: 4, dtype: str
```

```
[49]: if "gender" in df.columns:  
        df["gender"] = df["gender"].map({"Male": 1, "Female": 0, "Unknown/Invalid": -1})  
  
    if "diabetesMed" in df.columns:  
        df["diabetesMed"] = df["diabetesMed"].map({"Yes": 1, "No": 0})  
  
    if "change" in df.columns:  
        df["change"] = df["change"].map({"Ch": 1, "No": 0})  
  
    if "insulin" in df.columns:  
        print("Insulin unique values:", df["insulin"].unique())
```

```
Insulin unique values: <ArrowStringArray>  
['No', 'Up', 'Steady', 'Down']  
Length: 4, dtype: str
```

```
df.to_csv("dashboard_dataset.csv", index=False)
```

```
df = df.drop(columns=["readmitted"], errors="ignore")  
  
print("Object columns before encoding:")  
print(df.select_dtypes(include="object").columns)
```

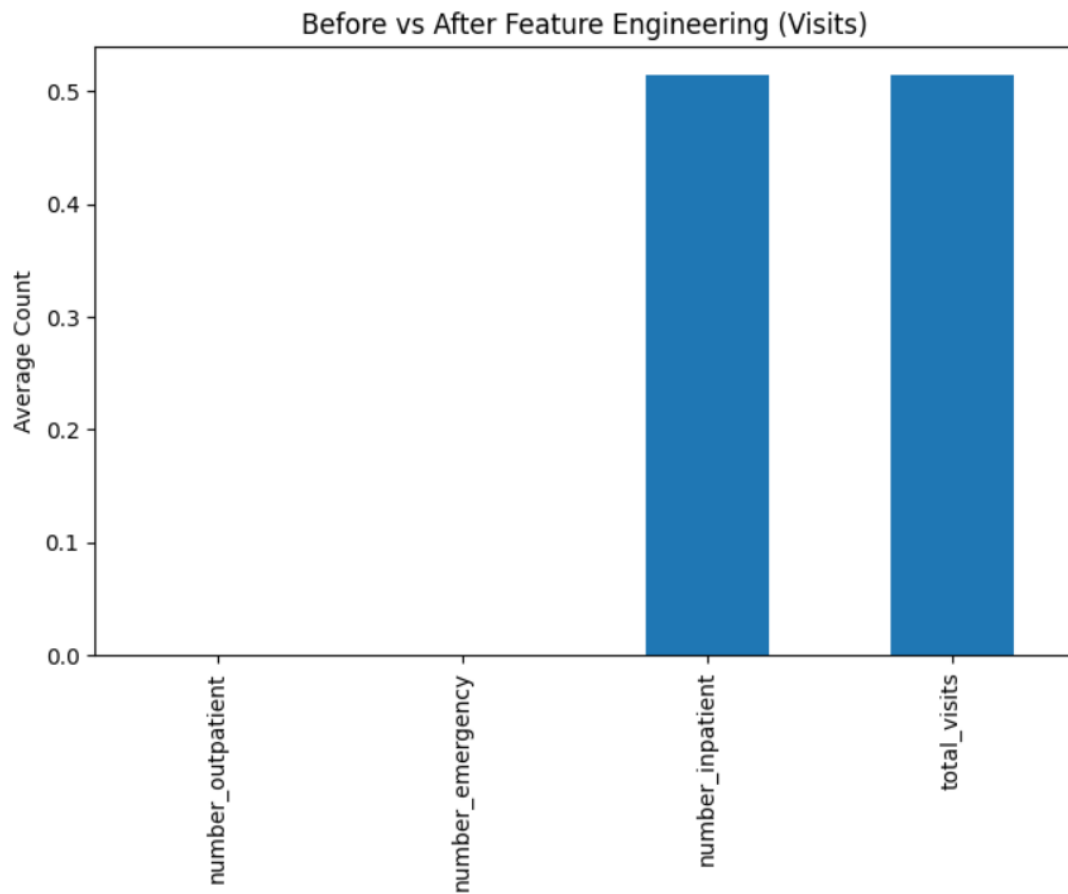
```
Object columns before encoding:  
Index(['race', 'age', 'medical_specialty', 'diag_1', 'diag_2', 'diag_3',  
       'max_glu_serum', 'A1Cresult', 'metformin', 'repaglinide', 'nateglinide',  
       'chlorpropamide', 'glimepiride', 'acetoexamide', 'glipizide',  
       'glyburide', 'tolbutamide', 'pioglitazone', 'rosiglitazone', 'acarbose',  
       'miglitol', 'troglitazone', 'tolazamide', 'examide', 'citoglipton',  
       'insulin', 'glyburide-metformin', 'glipizide-metformin',  
       'glimepiride-pioglitazone', 'metformin-rosiglitazone',  
       'metformin-pioglitazone'],  
      dtype='str')
```

```
plt.figure(figsize=(8,5))

df[['number_outpatient','number_emergency','number_inpatient','total_visits']].mean().plot(kind='bar')

plt.title("Before vs After Feature Engineering (Visits)")
plt.ylabel("Average Count")
plt.show()
```

OUTPUT:



ENCODING CATEGORICAL VARIABLES

```
] : le = LabelEncoder()
categorical_cols = df.select_dtypes(include="object").columns

for col in categorical_cols:
    df[col] = le.fit_transform(df[col].astype(str))

print(df.info())
```

OUTPUT:

```
0   race                101766 non-null   int64
1   gender              101766 non-null   int64
2   age                 101766 non-null   int64
3   admission_type_id   101766 non-null   int64
4   discharge_disposition_id 101766 non-null float64
5   admission_source_id 101766 non-null   int64
6   time_in_hospital    101766 non-null   int64
7   medical_specialty   101766 non-null   int64
8   num_lab_procedures  101766 non-null   int64
9   num_procedures       101766 non-null   int64
10  num_medications      101766 non-null   int64
11  number_outpatient     101766 non-null   int64
12  number_emergency      101766 non-null   int64
13  number_inpatient      101766 non-null float64
14  diag_1                101766 non-null   int64
15  diag_2                101766 non-null   int64
16  diag_3                101766 non-null   int64
17  number_diagnoses      101766 non-null float64
18  max_glu_serum         101766 non-null   int64
19  A1Cresult             101766 non-null   int64
20  metformin             101766 non-null   int64
21  repaglinide           101766 non-null   int64
22  nateglinide           101766 non-null   int64
23  chlorpropamide        101766 non-null   int64
24  glimepiride           101766 non-null   int64
25  acetohexamide         101766 non-null   int64
26  glipizide             101766 non-null   int64
27  glyburide             101766 non-null   int64
28  tolbutamide           101766 non-null   int64
29  pioglitazone           101766 non-null   int64
30  rosiglitazone         101766 non-null   int64
31  acarbose              101766 non-null   int64
32  miglitol              101766 non-null   int64
33  troglitazone          101766 non-null   int64
34  tolazamide            101766 non-null   int64
35  examide               101766 non-null   int64
36  citoglipton           101766 non-null   int64
37  insulin               101766 non-null   int64
38  glyburide-metformin   101766 non-null   int64
39  glipizide-metformin   101766 non-null   int64
40  glimepiride-pioglitazone 101766 non-null   int64
41  metformin-rosiglitazone 101766 non-null   int64
42  metformin-pioglitazone 101766 non-null   int64
43  change                101766 non-null   int64
44  diabetesMed           101766 non-null   int64
45  readmit_30            101766 non-null   int64
46  total_visits          101766 non-null float64
47  long_stay             101766 non-null   int64
```

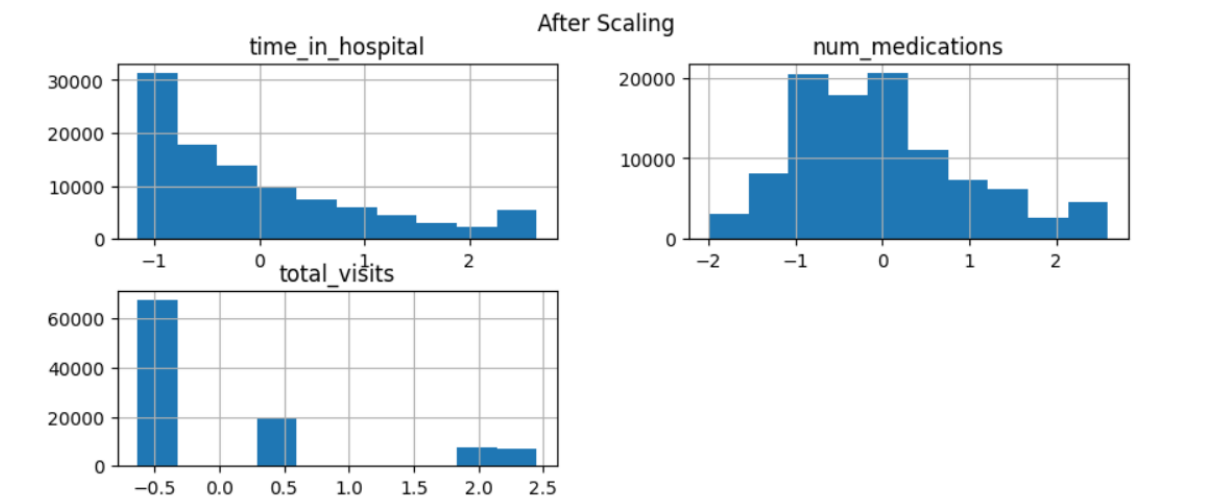
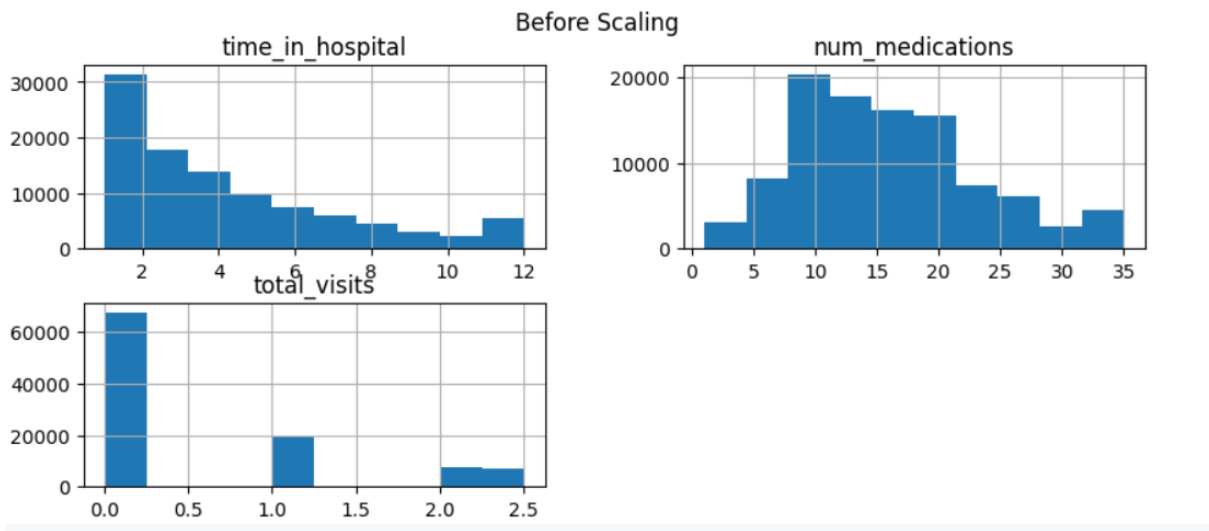
SCALING

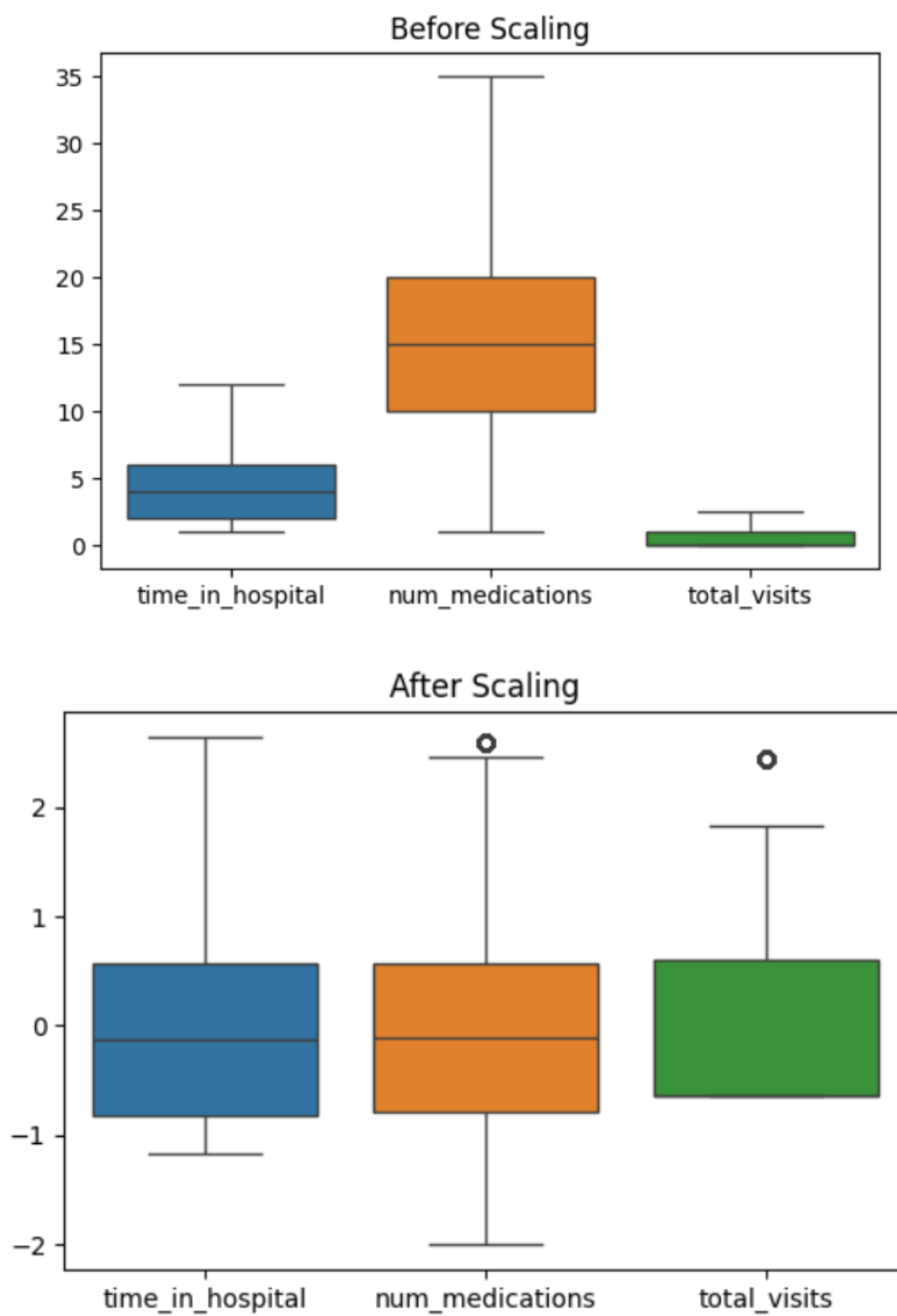
```
: scaler = StandardScaler()  
X_scaled = scaler.fit_transform(X_new)
```

```
: from sklearn.preprocessing import StandardScaler  
  
# Select features  
features = ['time_in_hospital', 'num_medications', 'total_visits']  
  
# Before scaling  
df[features].hist(figsize=(10,4))  
plt.suptitle("Before Scaling")  
plt.show()  
  
# Apply scaling  
scaler = StandardScaler()  
df_scaled = scaler.fit_transform(df[features])  
  
df_scaled = pd.DataFrame(df_scaled, columns=features)  
  
# After scaling  
df_scaled.hist(figsize=(10,4))  
plt.suptitle("After Scaling")  
plt.show()
```

```
# Before scaling  
plt.figure(figsize=(6,4))  
sns.boxplot(data=df[features])  
plt.title("Before Scaling")  
plt.show()  
  
# After scaling  
plt.figure(figsize=(6,4))  
sns.boxplot(data=df_scaled)  
plt.title("After Scaling")  
plt.show()
```

OUTPUT:





11/11/2024

FEATURE SELECTION

```
... X = df.drop("readmit_30", axis=1)
    y = df["readmit_30"]
```

```
... print(X.isnull().sum())
```

```
X = X.replace([np.inf, -np.inf], np.nan)
X = X.fillna(X.median(numeric_only=True))
```

```
X = X.fillna(X.mode().iloc[0])

print("Remaining missing values in X:", X.isnull().sum().sum())
print("Shape of X:", X.shape)
```

Remaining missing values in X: 0
Shape of X: (101766, 47)

```
from sklearn.ensemble import RandomForestClassifier
import pandas as pd
import matplotlib.pyplot as plt

X = df.drop('readmit_30', axis=1)
y = df['readmit_30']

model = RandomForestClassifier()
model.fit(X, y)

importances = model.feature_importances_
feat_importance = pd.Series(importances, index=X.columns)

feat_importance.nlargest(10).plot(kind='barh')
plt.title("Top Feature Importance")
plt.xlabel("Importance Score")
plt.show()
```

SELECT KBEST

```
: selector = SelectKBest(score_func=f_classif, k=10)
  X_new = selector.fit_transform(X, y)

  selected_features = X.columns[selector.get_support()]
  print("Selected features:", selected_features)
```

Selected features: Index(['discharge_disposition_id', 'time_in_hospital', 'num_lab_procedures',
'num_medications', 'number_inpatient', 'number_diagnoses', 'metformin',
'diabetesMed', 'total_visits', 'long_stay'],
dtype='str')

```

from sklearn.feature_selection import SelectKBest, f_classif

X = df.drop('readmit_30', axis=1)
y = df['readmit_30']

selector = SelectKBest(score_func=f_classif, k=10)
selector.fit(X, y)

scores = pd.Series(selector.scores_, index=X.columns)

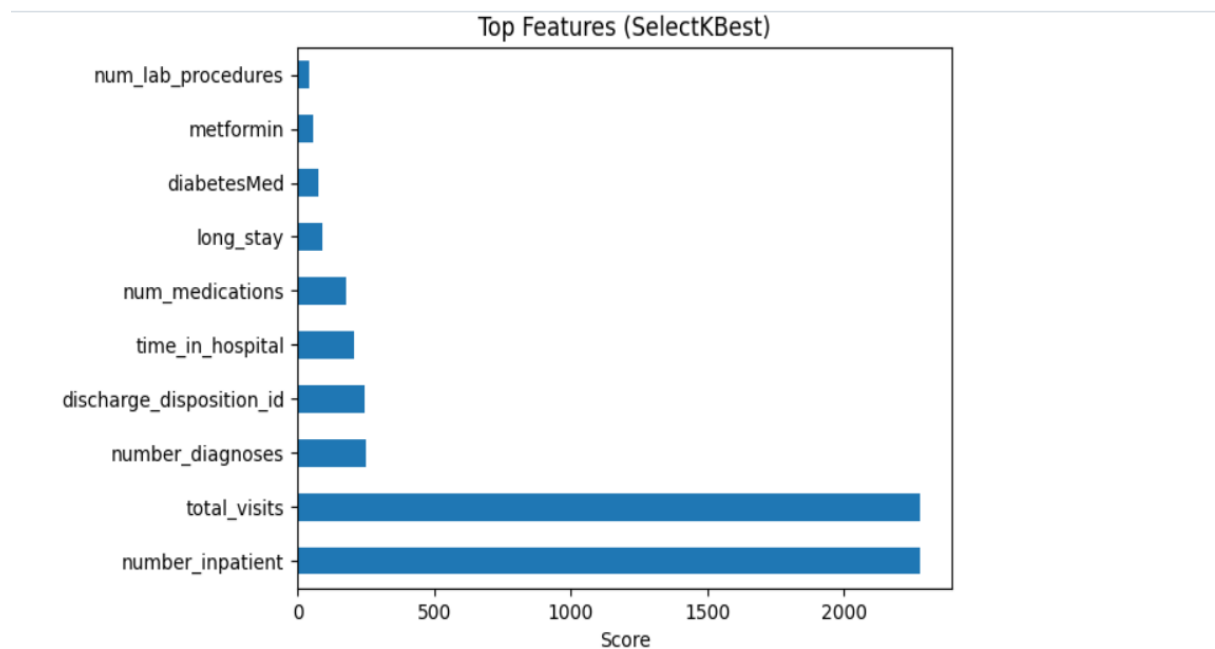
scores.nlargest(10).plot(kind='barh')
plt.title("Top Features (SelectKBest)")
plt.xlabel("Score")
plt.show()

```

OUTPUT:

race	0
gender	0
age	0
admission_type_id	0
discharge_disposition_id	0
admission_source_id	0
time_in_hospital	0
medical_specialty	0
num_lab_procedures	0
num_procedures	0
num_medications	0
number_outpatient	0
number_emergency	0
number_inpatient	0
diag_1	0
diag_2	0
diag_3	0
number_diagnoses	0
max_glu_serum	0
A1Cresult	0
metformin	0
repaglinide	0
nateglinide	0
chlorpropamide	0
glimepiride	0
acetohexamide	0
glipizide	0
glyburide	0
tolbutamide	0
pioglitazone	0
rosiglitazone	0
acarbose	0
miglitol	0
troglitazone	0
tolazamide	0
examide	0
citoglipton	0
insulin	0
glyburide-metformin	0
glipizide-metformin	0
glimepiride-pioglitazone	0
metformin-rosiglitazone	0
metformin-pioglitazone	0
change	0
diabetesMed	0
long_stay	0
total_visits	0

dtype: int64



PCA

```
]:  
pca = PCA()  
X_pca = pca.fit_transform(X_scaled)  
  
explained_variance = pca.explained_variance_ratio_  
  
plt.figure(figsize=(8, 5))  
plt.plot(np.cumsum(explained_variance), marker='o')  
plt.xlabel("Number of Components")  
plt.ylabel("Cumulative Explained Variance")  
plt.title("Explained Variance by PCA Components")  
plt.grid(True)  
plt.show()
```

```
pca = PCA(n_components=0.95)
X_pca = pca.fit_transform(X_scaled)

print("Original Shape:", X_scaled.shape)
print("Reduced Shape:", X_pca.shape)

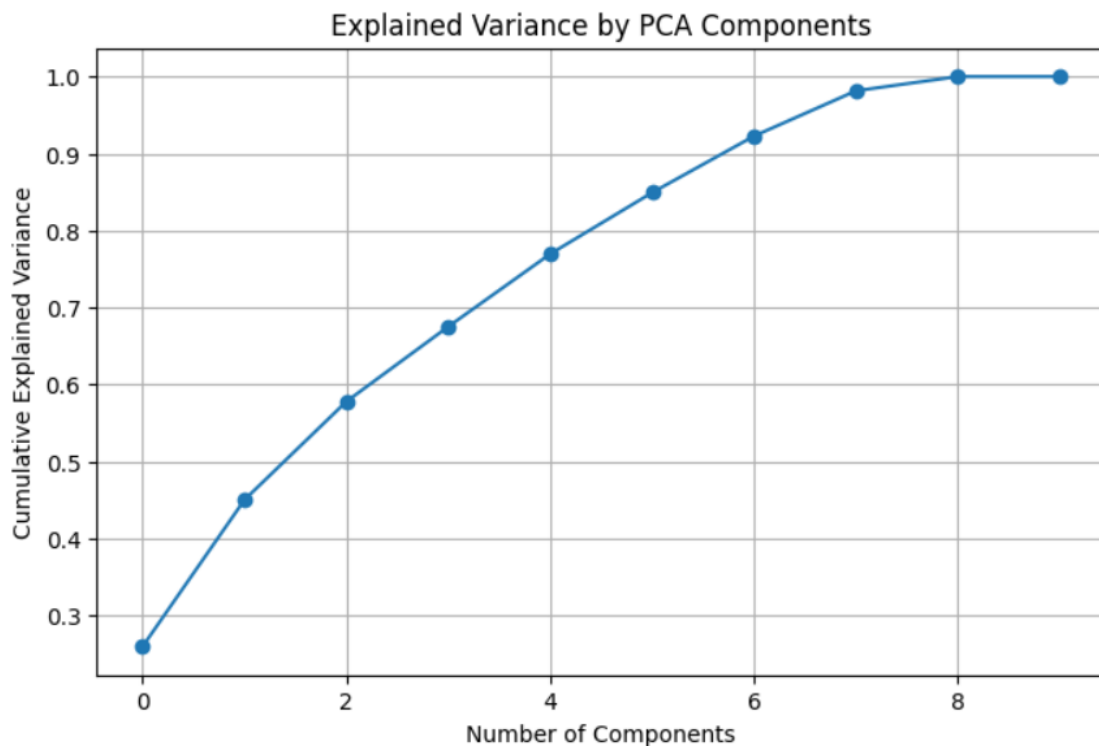
print("\nExplained Variance Ratio:")
print(pca.explained_variance_ratio_)
```

Original Shape: (101766, 10)
Reduced Shape: (101766, 8)

Explained Variance Ratio:
[0.25982164 0.19056901 0.12761463 0.09731611 0.09437216 0.0796721
0.07342015 0.05875827]

PCA VISUALIZATION

```
if X_pca.shape[1] >= 2:
    plt.figure(figsize=(8, 6))
    plt.scatter(X_pca[:, 0], X_pca[:, 1], alpha=0.5)
    plt.xlabel("Principal Component 1")
    plt.ylabel("Principal Component 2")
    plt.title("PCA Projection")
    plt.show()
else:
    print("PCA produced only 1 component, so scatter plot cannot be drawn.")
```



PCA WITH 95% VARIANCE

